

Aggregate Shocks and Cross-Section Dynamics: Quantifying Redistribution and Insurance in US Household Data

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Abstract

We use additive functionals and dynamic mode decompositions to analyze the co-evolution of cross-sections of private income, post-tax-and-transfer income, and consumption in the Consumer Expenditure Survey (CEX) from 1990 to 2023. After quantifying how cross-sectional inequality and redistribution interact with aggregate income, we construct value functions for quantiles of synthetic consumers who are exposed to both i.i.d. and serially correlated risks in income and consumption growth rates. For the median household, welfare costs from serially correlated risk are an order of magnitude larger than welfare costs from i.i.d. risk. For each quantile, we also compare benefits of eliminating risks in consumption growth with benefits from participating in the US tax-and-transfer system. In absolute values, benefits from the latter, which are positive (negative) for low (high) quantile consumers, far exceed those from the former.

Keywords: Long-run risk, welfare, heterogeneity, common trends, business cycles.

JEL Classification: E3, E6, C7

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1 Introduction

We use a parametric statistical model of cointegrated CEX cross-section consumption and income quantiles to organize evidence about how insurance and redistribution across quantiles interact with aggregate income.¹ We infer how aggregate shocks affect incomes at different quantiles. We use our statistical model to construct value functions that quantify how much insurance and redistribution the US tax-and-transfer system provides. We find magnitudes of effects of the tax-and-transfer system that far exceed those from further stabilizing the business cycle, especially for lower-income households. We confirm data patterns that rationalize representative agent macroeconomic models and an associated “neoclassical synthesis” that separates optimal monetary-fiscal policies to moderate aggregate business cycles from optimal microeconomic policies that redistribute income and insure against idiosyncratic shocks.²

Our statistical model is what [Koopmans \(1947\)](#) called a “Kepler-stage” descriptive model of business cycles to distinguish it from the “Newton-stage” structural models that Koopmans and his Cowles Commission colleagues used to study consequences of historically unprecedented government policies.³ [Koopmans](#) interpreted [Burns and Mitchell’s \(1946\)](#) book as presenting an algorithm that they had used to infer a single business cycle “factor” that intermediates co-movements of a diverse set of quantities and nominal prices.⁴ We synthesize an aggregate (earned) income series as a cross-section mean of private income from CEX quantiles.⁵ We posit that this aggregate income series is an additive functional that is driven by a first-order vector autoregression for a vector consisting of aggregate income growth and quantiles of the three cross-sections scaled by aggregate income. We use ideas from [Hansen \(2012\)](#) to decompose aggregate income into deterministic trend, martingale,

¹These CEX consumption and post-tax-and-transfer income cross-sections reflect monetary-fiscal policies aimed at stabilizing business cycles and fiscal policies that redistribute and insure. This is important when it comes to interpreting our Section 5 welfare comparisons.

²For accounts of the “neoclassical synthesis” and its origins, see [Lucas \(1987\)](#) and [Sargent \(2015, 2024, 2025\)](#). For recent evidence about this issue that uses a different data source and a different statistical model than we do, see [Bilbiie et al. \(2025\)](#).

³See [Koopmans \(1950\)](#), [Hood and Koopmans \(1953\)](#), and [Marschak \(1953\)](#).

⁴As emphasized by [Koopmans \(1947\)](#) and [Sargent and Sims \(1977\)](#), the role of a purely descriptive parametric statistical model like ours is to detect and organize patterns that should be matched by a “structural” statistical model, cast in terms of parameters that can be interpreted as describing purposes and constraints of economic decision makers who live inside the model.

⁵Thus, our measure of this “aggregated” variable emerges from measurements of diverse disaggregated variables, in the tradition of [Burns and Mitchell](#). We use no additional data on aggregate quantities.

and stationary components and to describe how aggregate income interacts with CEX cross-sections of private income, post-tax income, and consumption.⁶ Our specification implies that quantiles for these three CEX cross sections are also additive functionals that can be decomposed into their trend, martingale, and stationary components.⁷ We use these representations to quantify interactions between aggregate income and the three CEX cross-sections. We also use them to construct quantile-specific value functions that allow us to quantify how the US tax and transfer system redistributes income across quantiles and insures different quantiles against shocks to income growth rates.

Section 2 describes how we construct quantiles for the CEX data. Section 3 describes how we construct additive functionals. This section also explains how we purposefully designed our statistical model to facilitate our Section 5 welfare calculations. Section 4 describes dynamic mode decompositions and their connection to vector autoregressions. It also presents findings about how aggregate income influences cross-sections. Section 5 constructs welfare comparisons like those of Lucas (1987, Sec. III) and Lucas (2003), who measured the welfare of a representative consumer exposed to i.i.d. risk in consumption growth. We extend Lucas’s analysis to study a cross section of consumers who are exposed to serially correlated risks in consumption growth. We compare how different quantiles would benefit from increasing the rate of growth of log consumption and from eliminating exposures to consumption growth risk. We also study welfare consequences from participating in the existing US tax-and-transfer system. Section 6 concludes. Appendices provide technical details.

2 Data Description and Compression

The Consumer Expenditure Survey (CEX) is a representative survey of US households conducted by the Bureau of Labor Statistics. We study quarterly CEX waves from 1990Q1 to 2023Q1 ($T = 133$ observations). Our study covers three variables: (1) Private income –

⁶By basing our statistical representations on decompositions of additive functionals described by Hansen (2012), we avoid pitfalls described by Hamilton (2018) that are associated with the Hodrick-Prescott pre-filtering method that is still widely used in macroeconomics.

⁷Satisfaction of technical conditions described by Sargent et al. (2025) justifies our use of a first-order VAR rather than the higher-order VAR that appears in the more general additive functional of Hansen (2012). Hansen (2012) is formulated in continuous time; for a discrete-time exposition, see the QuantMFR lecture Borovička et al. (2025) and the QuantEcon lecture Sargent and Stachurski (2025) on additive and multiplicative functionals.

labor income plus financial income, (2) Post-tax-and-transfer income – private income plus transfers minus taxes, and (3) Consumption.⁸

For each time t , we rank households by their real consumption levels, which we construct by deflating by the PCE chain-type price index. Let $\hat{c}_{i,t}$ denote the real consumption of household $i = 1, \dots, N_t$ at time t , where N_t is the number of households observed at time t . We partition the ranked households into 100 bins. For each bin $p = 0, 1, \dots, 99$, let $\mathcal{P}_{p,t}$ denote the set of households in that bin at time t . If N_t is divisible by 100, each set contains exactly $N_t/100$ households. Otherwise, the first $p = 0, \dots, 98$ sets each contain $\lfloor N_t/100 \rfloor$ households, and the $p = 99$ set contains the remainder plus $\lfloor N_t/100 \rfloor$ households. We define the log real consumption of bin p as

$$c_{p,t} = \log \left(\frac{1}{N_{p,t}} \sum_{i \in \mathcal{P}_{p,t}} \hat{c}_{i,t} \right), \quad p = 0, \dots, 99,$$

where $N_{p,t}$ denotes the number of households in bin p at time t . Note that $c_{p,t}$ is the log of the mean consumption of households in bin p . For brevity, we use the term “ p th percentile” interchangeably with “ p th bin average” throughout this paper, with the understanding that it refers to these household groups representing an average household in each bin, rather than statistical percentiles.

We define corresponding log real private-income and log real post-tax income bins analogously and denote them by $m_{p,t}$ and $d_{p,t}$ using the same procedure.⁹ We also deflate them by the PCE chain-type price index. We keep $M = 100$ bins indexed by $p \in \{0, 1, \dots, 99\} =: \mathcal{P}$ for each variable. The vectors $\mathbf{m}_t = [m_{0,t}, \dots, m_{99,t}]^\top$, $\mathbf{d}_t = [d_{0,t}, \dots, d_{99,t}]^\top$ and $\mathbf{c}_t = [c_{0,t}, \dots, c_{99,t}]^\top$ collect the bins in $(M \times 1)$ column vectors for private- and post-tax income and consumption respectively.

For each time period t , we define a Chisini mean (Chisini, 1929) of the cross-section of $\mathbf{u}_t \in \{\mathbf{m}_t, \mathbf{d}_t, \mathbf{c}_t\}$ as

$$Y_t^{(u)} = \frac{1}{M} \sum_{p \in \mathcal{P}} u_{p,t}. \quad (1)$$

⁸See Appendix A for details on the data construction and preparation.

⁹For each of consumption, private income, and post-tax income we rank households separately, so a household’s consumption bin need not coincide with its bins in the income distributions.

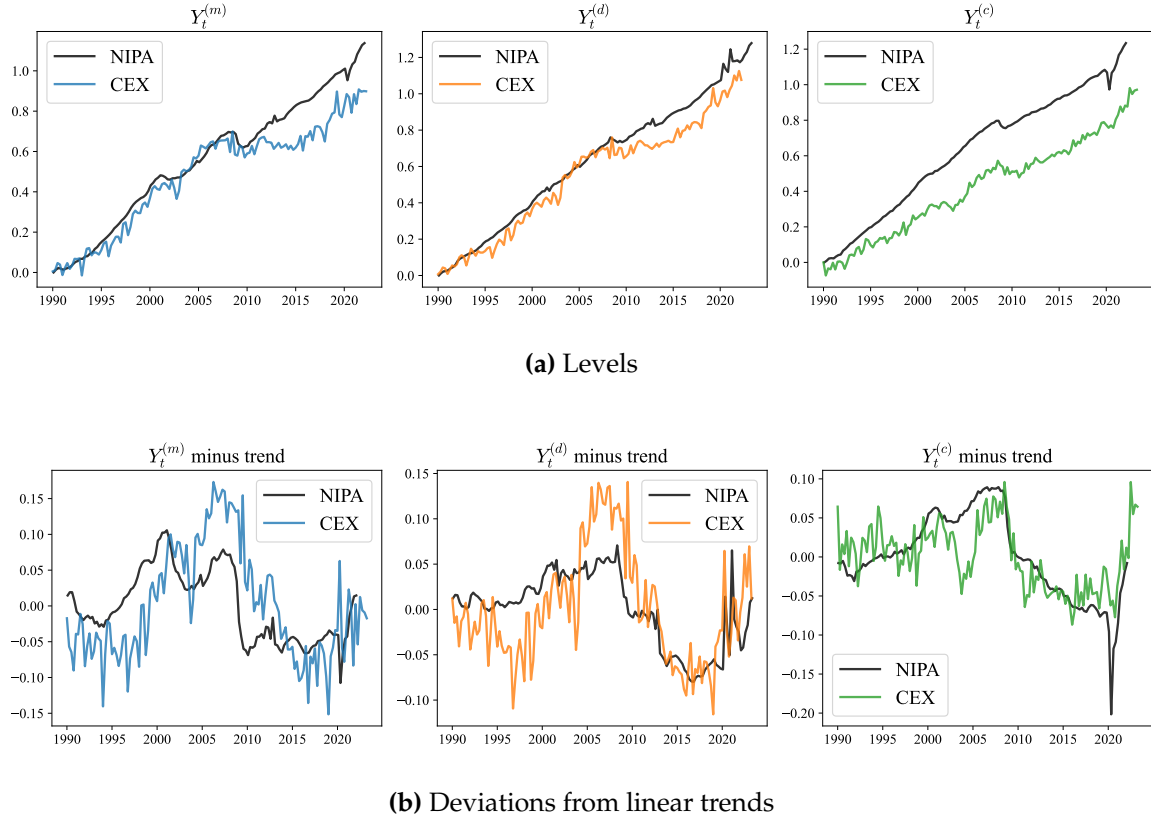


Figure 1: NIPA and Chisini CEX means

Figure 1 compares nominal versions of our Chisini means from the CEX quantiles with corresponding aggregates from the National Income and Product Accounts (NIPA). Here $Y_t^{(m)}$, $Y_t^{(d)}$, and $Y_t^{(c)}$ denote Chisini means of private income, post-tax income, and consumption, respectively.¹⁰ Levels of the series differ in panel (a), especially for consumption, but their cyclical patterns in panel (b) are broadly similar.¹¹ Differences in consumption levels can be attributed to coverage and definition differences. NIPA consumption includes services to nonprofit institutions, government expenditures (Medicare/Medicaid), employer expenditures, owner-occupied rent, and financial services and insurance, which are not included in

¹⁰We compute private income using NIPA Table 2.1 on Personal Income and its Disposition. NIPA private income = Personal income (1) - (Employer) supplements to wages and salaries (6) - Government social benefits (17) + Contributions to government social insurance (25). The numbers in parentheses indicate the line item in NIPA Table 2.1.

¹¹Let Y_t denote the per capita series. We compute the linear trend τ_t

$$\hat{\nu} := \frac{1}{T-1} \sum_{t=2}^T (Y_t - Y_{t-1}), \quad \tau_t := Y_0 + \hat{\nu}t.$$

Define the gap $g_t := Y_t - \tau_t$ and denote its sample mean by $\bar{g}$. Panel (b) plots the demeaned gaps $g_t - \bar{g}$ for NIPA and CEX.

the CEX measure. According to [Carroll et al. \(2015\)](#), these differences account for somewhat more than a quarter of the gap between the two series from 1992 to 2010.

3 Kepler-Stage Descriptive Statistical Model

3.1 Choice of Approximating Model

We designed our statistical model with our eyes on the Section 5 value functions that we construct to extend welfare calculations of [Lucas \(1987, 2003\)](#). Proposition 2 in Section 5 tells how those value functions satisfy Bellman equations that can be solved by applying operators presented by [Hansen et al. \(2008\)](#) and [Hansen and Sargent \(2010\)](#). Those Bellman equations assume that the statistical model underlying our calculations takes the form that we describe in this section. We have further specialized our statistical model by assuming that the vector autoregression that appears in our additive functional is a first-order reduced rank VAR. Doing this lets us connect our analysis to a dynamic mode decomposition (DMD) and thereby avail ourselves of the relationship between reduced-rank first-order VARs and DMDs analyzed by [Sargent et al. \(2025\)](#).

While this approach allows us to construct our Section 5 welfare comparisons, it has costs by making the model a poor approximation to some aspects of our data. Thus, because our statistical model asserts that conditional distributions of future values of the quantiles of the log income and log consumption cross sections are multivariate normal, they cross with positive probabilities, which of course they can't in reality. Unrealistic behavior like this is something users of a linear-quadratic-Gaussian model like ours just have to accept.¹²

3.2 Our Statistical Model

We cast our statistical model in terms of the following objects. Recall that $\mathbf{m}_t, \mathbf{d}_t, \mathbf{c}_t \in \mathbb{R}^M$ are vectors of quantiles for private income, post-tax income, and consumption at time t . For each $t \in \{1, \dots, T\}$, we subtract the Chisini mean of private income $Y_t := Y_t^{(m)}$ from all

¹²The LQG dynamic stochastic general equilibrium models of [Hansen and Sargent \(2013\)](#) share such problems. Thus, equilibrium prices in those models can be negative, though a clever user can make negative prices occur with only small probabilities. [Sargent et al. \(2025\)](#) use a [Hansen and Sargent \(2013, ch. 12\)](#) heterogeneous-agent general equilibrium model as a laboratory for studying DMD as an approximating statistical model. See footnote 16 below.

cross-section quantiles¹³:

$$\tilde{\mathbf{m}}_t = \mathbf{m}_t - Y_t \mathbf{1}, \quad (2)$$

$$\tilde{\mathbf{d}}_t = \mathbf{d}_t - Y_t \mathbf{1}, \quad (3)$$

$$\tilde{\mathbf{c}}_t = \mathbf{c}_t - Y_t \mathbf{1}, \quad (4)$$

where $\mathbf{1} \in \mathbb{R}^M$ is a vector of ones.

Let $\boldsymbol{\nu}^{(i)} \in \mathbb{R}^M$ denote the vector of time-series means for each quantile for $i \in \{m, d, c\}$, and let $\nu := \frac{1}{T-1} \sum_{t=2}^T (Y_t - Y_{t-1}) \in \mathbb{R}$ denote the time series mean of aggregate income growth. We define $\mathbf{y}_t \in \mathbb{R}^{3M+1}$ as

$$\mathbf{y}_t = \begin{bmatrix} Y_t - Y_{t-1} - \nu \\ \tilde{\mathbf{m}}_t - \boldsymbol{\nu}^{(m)} \\ \tilde{\mathbf{d}}_t - \boldsymbol{\nu}^{(d)} \\ \tilde{\mathbf{c}}_t - \boldsymbol{\nu}^{(c)} \end{bmatrix}. \quad (5)$$

We assume that aggregate private income $\{Y_t\}$ is an additive functional driven by the following first order vector autoregression:

$$\mathbf{y}_{t+1} = \mathbf{B} \mathbf{y}_t + \mathbf{a}_{t+1} \quad (6)$$

$$Y_{t+1} - Y_t - \nu = e_1 \mathbf{B} \mathbf{y}_t + e_1 \mathbf{a}_{t+1}, \quad (7)$$

where $\mathbf{B} \in \mathbb{R}^{(3M+1) \times (3M+1)}$ is a transition matrix with spectral radius $\rho(\mathbf{B}) < 1$, $\mathbf{a}_{t+1} \in \mathbb{R}^{3M+1}$ are innovations satisfying $\mathbb{E}[\mathbf{a}_{t+1}] = \mathbf{0}$, $\mathbb{E}[\mathbf{a}_{t+1} \mathbf{y}_{t-j}^\top] = \mathbf{0}$ for all $j \geq 0$, $\mathbb{E}[\mathbf{a}_{t+1} \mathbf{a}_{t+1}^\top] = \boldsymbol{\Omega}$, and $e_1 = [1 \ 0 \ \dots \ 0] \in \mathbb{R}^{1 \times (3M+1)}$ is a row vector that selects the first entry of $\mathbf{y}_t$, namely $Y_t - Y_{t-1} - \nu$. We describe a dynamic mode decomposition algorithm for estimating $\mathbf{B}$ in Section 4.

Following Hansen (2012)¹⁴, we decompose Y_t into its trend, martingale, stationary, and constant components.

¹³By construction, the cross-sectional mean $\frac{1}{M} \sum_{p \in \mathcal{P}} \tilde{m}_{p,t} = 0$ for all t . We compute cross quantile means and standard deviations for each variable. For example, for private income we compute $\mu_t^{(m)} := \frac{1}{M} \sum_{p \in \mathcal{P}} \tilde{m}_{p,t}$ and $\sigma_t^{(m)} := \sqrt{\frac{1}{M} \sum_{p \in \mathcal{P}} (\tilde{m}_{p,t} - \mu_t^{(m)})^2}$. We report these moments in Figure 9 in Appendix A.

¹⁴Also see the QuantEcon lecture on additive and multiplicative functionals available in Sargent and Stachurski (2025).

Proposition 1 (Additive Functional Decomposition). *Suppose $\mathbf{y}_t$ and Y_t satisfy (6)-(7) and spectral radius $\rho(\mathbf{B}) < 1$. Then Y_t has representation*

$$Y_t = \underbrace{t\nu}_{\text{Trend}} + \underbrace{\sum_{j=1}^t \mathbf{H} \mathbf{a}_j}_{\text{Martingale}} \underbrace{-\mathbf{g} \mathbf{y}_t}_{\text{Stationary}} + \underbrace{(\mathbf{g} \mathbf{y}_0 + Y_0)}_{\text{Constant}}. \quad (8)$$

where $\mathbf{H}, \mathbf{g} \in \mathbb{R}^{1 \times (3M+1)}$ are defined by

$$\mathbf{H} = e_1 + e_1 \mathbf{B}(\mathbf{I} - \mathbf{B})^{-1}, \quad (9)$$

$$\mathbf{g} = e_1 \mathbf{B}(\mathbf{I} - \mathbf{B})^{-1}. \quad (10)$$

Proof. See Appendix B.1. □

The four components in (8) are: (i) a deterministic trend $t\nu$, (ii) a martingale $\sum_{j=1}^t \mathbf{H} \mathbf{a}_j$, (iii) an asymptotically stationary component $-\mathbf{g} \mathbf{y}_t$, and (iv) a constant $(\mathbf{g} \mathbf{y}_0 + Y_0)$.¹⁵

Figure 2 presents decomposition (8) under two alternative estimation samples. The left panel uses all model parameters $\nu, \nu^{(m)}, \nu^{(d)}, \nu^{(c)}, \mathbf{B}, \boldsymbol{\Omega}$ from CEX data through 2008Q4 and then applies decomposition (8) to the full 1990–2023 sample. The right panel estimates all parameters from the full 1990–2023 sample. The left panel provides a baseline case that uses pre-crisis parameters to characterize subsequent developments, thereby quantifying how much actual outcomes departed from the pre-crisis deterministic trend. In the remainder of this section, we use the pre-crisis parameter estimates shown in the left panel because they are associated with a deterministic linear trend value used in much macro literature that describes how post-crisis U.S. macro aggregates have remained below trend since 2008. Sections 4 and 5 use the full-sample parameter estimates shown in the right panel to account for post-crisis dynamics and their welfare consequences.

In the left panel, $t\nu$ is our estimate of the “pre-crisis” deterministic trend. Shortly before the 2008 financial crisis, the martingale component was close to zero and Y_t was above its deterministic trend. After 2008, Y_t fell and remained persistently below deterministic trend until the end of the sample in 2023. Initially, the martingale component dropped to minus

¹⁵The martingale and stationary components are typically correlated.

15%. The stationary component declined gradually and became negative in 2014Q4, while the martingale component increased until 2020. The fall in Y_t during 2021 came mostly from the stationary component. At the end of the sample, Y_t was around 20% below trend. The right panel shows that when parameters are estimated from the full sample, the deterministic trend is lower (reflecting slower post-2008 growth). There was substantial heterogeneity in quantile-specific stationary components after 2008, which we now proceed to describe.

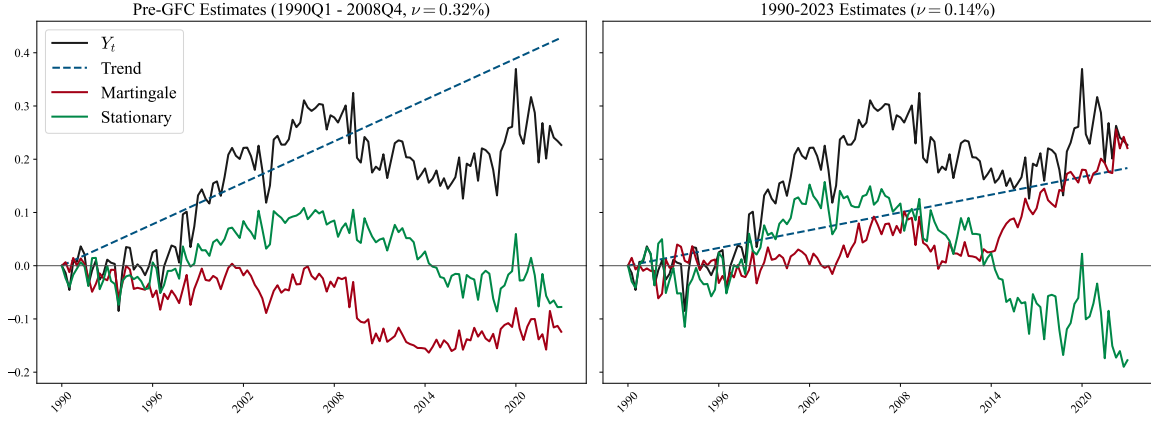


Figure 2: Decomposition of $Y_t - Y_0$ into trend, martingale, and stationary components. Left panel: parameters estimated from 1990Q1–2008Q4 (pre-crisis), then applied to the full sample. Right panel: parameters estimated from the full 1990–2023 sample.

Corollary 1 (Quantile-Specific Decompositions). *Under the conditions of Proposition 1, each quantile $m_{p,t}$, $d_{p,t}$, $c_{p,t}$ for $p \in \mathcal{P}$ admits a decomposition into trend, martingale, stationary, and constant components. In particular, for each $u \in \{m, d, c\}$ and $p \in \mathcal{P}$ there exist row vectors $e_p^{(u)} \in \mathbb{R}^{1 \times (3M+1)}$ and scalars $\nu_p^{(u)} \in \mathbb{R}$ such that*

$$u_{p,t} = \underbrace{t\nu}_{\text{Trend}} + \underbrace{\sum_{j=1}^t \mathbf{H} \mathbf{a}_j}_{\text{Martingale}} - \underbrace{(\mathbf{g} - e_p^{(u)}) \mathbf{y}_t}_{\text{Stationary}} + \underbrace{(\mathbf{g} \mathbf{y}_0 + Y_0 + \nu_p^{(u)})}_{\text{Constant}}. \quad (11)$$

Proof. See Appendix B.2. □

Notice that while decompositions (8) and (11) share common deterministic trend $t\nu$ and martingale $\sum_{j=1}^t \mathbf{H} \mathbf{a}_j$ components across all quantiles and variables, they have quantile-specific stationary and constant components. This structure is built into our statistical model

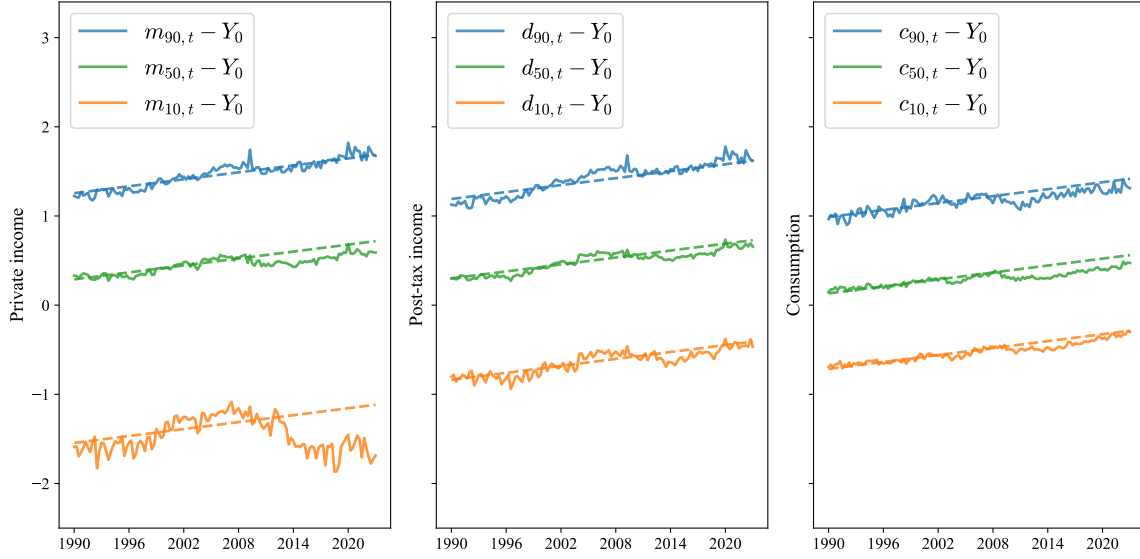


Figure 3: Evolution of quantiles. The dashed line represents pre-GFC trend.

specification (6)–(7).

To create Figure 3, we have used appropriate versions of (11) to decompose income and consumption quantiles. For the 10th percentile, private income is much lower than both post-tax income and consumption. For 90th percentile households, private incomes are higher than post-tax incomes, indicating substantial transfers from higher quantiles of the distribution. Differences across percentiles are largest for private income and smallest for consumption.¹⁶

The deviations of quantiles from their deterministic trend in Figure 3 indicate interesting patterns. After the 2008 financial crisis, private income of the 90th percentile stayed close to or above its pre-crisis trend throughout the sample. Similar patterns apply to 90th percentile post-tax income, but not to consumption. After 2008, 90th percentile consumption fell below deterministic trend and stayed there through the end of our sample. As for lower quantiles,

¹⁶It is enlightening to compare Figure 3 with Figure 3 in Section 3 of Sargent et al. (2025), which plots realizations of three series from a competitive equilibrium of a complete markets, heterogeneous-agent linear quadratic general equilibrium model that satisfies conditions for Gorman aggregation described in Hansen and Sargent (2013, ch. 12). The left panel of their Figure 3 plots quantiles of private income before taxes and transfers. The right panel plots consumption quantiles after state-contingent, Pareto efficient taxes and transfers that can be described by the post-tax-and-transfer income quantiles described in the middle panel. Patterns in Sargent et al.'s Figure 3 remind us of those in Figure 3 in this paper, with the notable difference that Sargent et al.'s general equilibrium model exhibits no deterministic trend growth, a feature that they intentionally built in to serve their purpose of creating a lab within which to understand how their DMD statistical model could serve as a statistical approximation to outcomes produced by a heterogeneous agent DSGE model with ample social insurance. To do that, they excluded deterministic trends and adopted additional assumptions that imply stationary prices and quantities.

after 2008, private incomes at the 10th percentile fell substantially below trend and exhibited high volatility. By the end of the sample, 10th percentile private income was about 50 percent below trend. Post-tax income and consumption for these households, however, almost tracked deterministic trend throughout the sample.

A striking aspect of Figure 3 is that, at the 10th percentile, post-tax income is systematically higher and less volatile than private income, while post-tax income and consumption are very similar. This indicates substantial redistribution and insurance at the 10th percentile, but much weaker consumption smoothing through management of private savings. Things look very different at the 90th percentile, where post-tax income is systematically lower than private income, and consumption is systematically lower than post-tax income. This indicates the presence of substantial redistribution and social insurance and also of consumption smoothing through management of private savings. These differences across quantiles shape differences in discounted expected utilities that we report in Section 5.

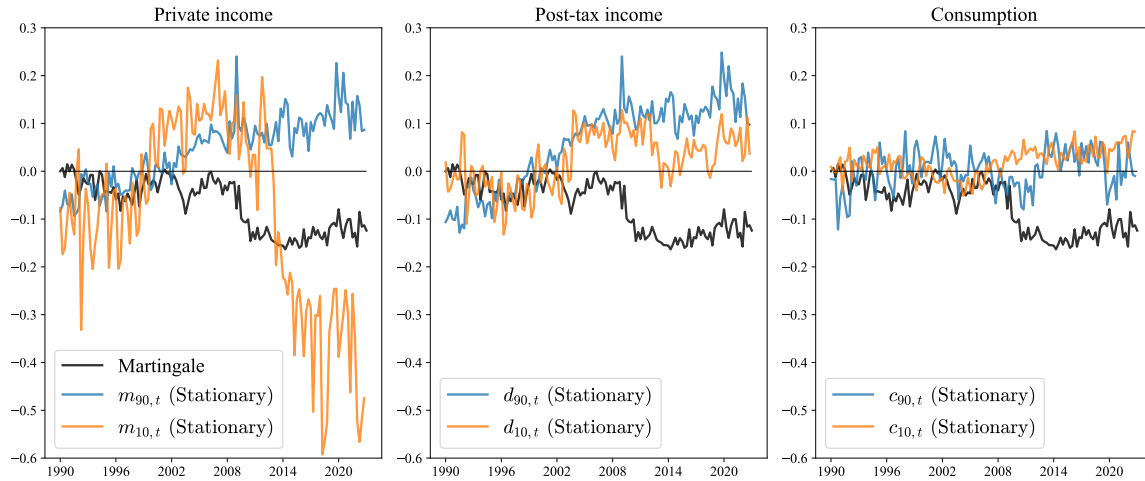


Figure 4: Components of quantiles

Figure 4 provides another perspective by using equation (11) to decompose deviations of quantiles from their trends into two parts: a common martingale component of Y_t (black) that makes the same contributions to all quantiles of all three variables, and idiosyncratic stationary components (blue for 90th percentiles, orange for 10th percentiles). As plotted in all three panels of Figure 4, the common martingale component fell 15% during the 2008 crisis and gradually recovered to about minus 10% at the end of our sample.

The left panel of Figure 4 shows that for the 10th percentile, martingale and stationary components both contribute to the large drop in private incomes relative to trend. While the martingale component dominates initially, outcomes after 2014 are mostly driven by the idiosyncratic stationary component, which accounts for the entire gap from deterministic trend by 2023. On the other hand, after 2008 the stationary component of the 90th percentile was slightly positive; by the end of the sample, it mostly offset the negative martingale component.

Turning to the middle panel of Figure 4, post-tax income stationary components for the 10th percentile were close to zero, making the common martingale component the predominant driver of the fall relative to trend. The stationary component for the 90th percentile was positive after 2008 and therefore partially offset the negative martingale component.

Turning now to consumption displayed in the right panel of Figure 4, stationary components of both 10th and 90th percentiles were close to zero, so that the common martingale component made the most important contribution to consumption. The idiosyncratic stationary components are, on the whole, close to zero for the 10th and 90th consumption percentiles, indicating pervasive risk-sharing.¹⁷

4 Dynamic Mode Decomposition

We now discuss a dynamic mode decomposition (DMD) and the associated reduced rank first-order VAR that we used to estimate parameters of our statistical model (6)-(7). In this section, we use parameter estimates from the full 1990–2023 sample (the right-panel estimates in Figure 2). In Section 4.1 we tell how we applied a dimension reduction technique based on a singular value decomposition that underlies a DMD. In Section 4.2 we extract quantitative inferences about interactions between aggregate income and our cross-section quantiles. Appendix C describes the DMD algorithm we used to estimate parameters while Appendix

¹⁷This feature is consistent with the bottom right panel of Figure 11 that plots time-series of demeaned $\tilde{c}_{p,t}$ for different percentiles. It shows that consumption paths are highly correlated across quantiles, exhibiting very similar fluctuations in our sample. In macro models with preferences and complete market structures like those presented in [Ljungqvist and Sargent \(2018, ch. 8\)](#), highly correlated consumption paths like these indicate the presence of widespread sharing of aggregate risks. [Sargent et al. \(2025, sec. 4\)](#) describe a complete markets economy with heterogeneous agents that generates such correlated consumption paths through efficient risk-sharing arrangements.

D tells why we decided to retain three modes in estimating our statistical model (6)-(7), describes estimates of parameters governing the dynamics of those three modes, and then plots three dominant DMD modes against some components of our data set.

4.1 Estimation Details

We estimated parameters of the VAR (6) by constructing a dynamic mode decomposition (DMD), a “machine-learning” technique used to study fluid dynamics¹⁸ that Sargent et al. (2025) connect to a first-order vector autoregression and an associated linear Gaussian state-space system. Our analysis begins with a data set formed by quarterly observations of the vector $\mathbf{y}_t \in \mathbb{R}^{(3M+1) \times 1}$ defined in (5). We use these data to estimate the first-order VAR

$$\begin{aligned} \mathbf{y}_{t+1} &= \mathbf{B} \mathbf{y}_t + \mathbf{a}_{t+1} \\ \mathbb{E}[\mathbf{a}_{t+1} \mathbf{a}_{t+1}^\top] &= \mathbf{\Omega}, \quad \mathbf{a}_{t+1} \perp \mathbf{y}_t \end{aligned} \tag{12}$$

The coefficient matrix $\mathbf{B}$ has dimension $(3M + 1) \times (3M + 1)$ which, since $M = 100$, is 301×301 in our application. Our 33 years of quarterly observations mean that we have $T = 133$ observations with which to estimate the 301^2 coefficients in $\mathbf{B}$. Thus, specification (6) confronts us with an underdetermined least squares problem that we solve by using ideas connected to DMD. We form the data matrices

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \cdots & \mathbf{y}_{T-1} \end{bmatrix}, \quad \mathbf{Y}' = \begin{bmatrix} \mathbf{y}_2 & \mathbf{y}_3 & \cdots & \mathbf{y}_T \end{bmatrix}$$

where $\mathbf{Y}, \mathbf{Y}' \in \mathbb{R}^{(3M+1) \times (T-1)}$. Figure 5 plots the 50 largest singular values of the DMD data matrix $\mathbf{Y}$. We set $N = 3 \ll T$ via the “elbow rule,” truncating where the singular value decay levels off. We then seek a rank- N first-order VAR representation

$$\begin{aligned} \mathbf{y}_{t+1} &= \hat{\mathbf{B}} \mathbf{y}_t + \hat{\mathbf{a}}_{t+1} \\ \mathbb{E}[\hat{\mathbf{a}}_{t+1} \hat{\mathbf{a}}_{t+1}^\top] &= \hat{\mathbf{\Omega}}, \quad \hat{\mathbf{a}}_{t+1} \perp \mathbf{y}_t. \end{aligned} \tag{13}$$

¹⁸See, for example, Tu et al. (2014) and Brunton and Kutz (2022).

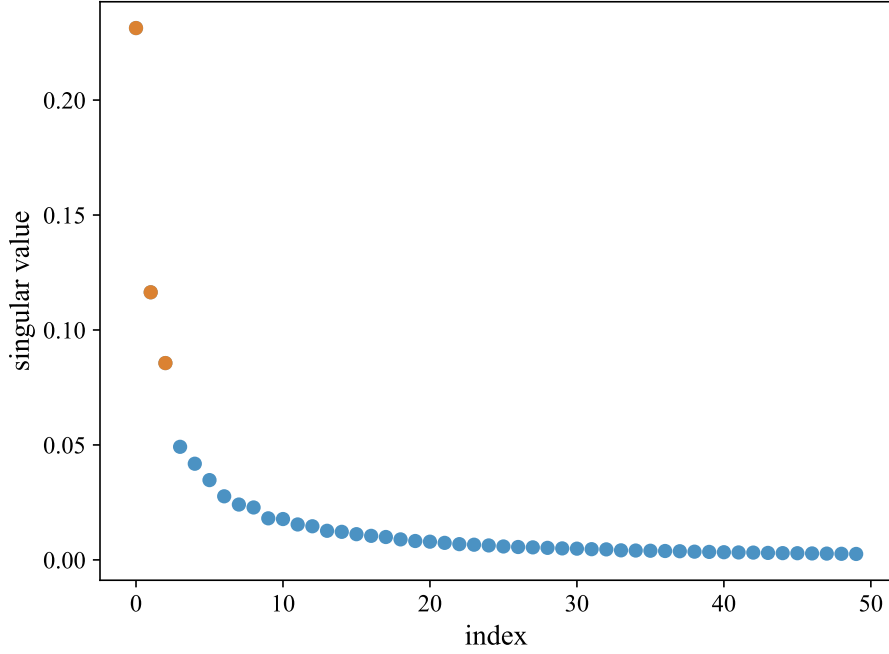


Figure 5: 50 largest singular values of DMD data matrix $\mathbf{Y}$

where $\hat{\mathbf{B}}$ is a rank N matrix and

$$\hat{\mathbf{a}}_{t+1} := \mathbf{y}_{t+1} - \hat{\mathbf{B}} \mathbf{y}_t, \quad \hat{\mathbf{\Omega}} := \frac{1}{T-1} \sum_{t=1}^{T-1} \hat{\mathbf{a}}_{t+1} \hat{\mathbf{a}}_{t+1}^\top.$$

Appendix [D](#) defines DMD “modes”, describes how we selected N in detail, presents our estimates of $\mathbf{\Lambda}$ as well as our three DMD modes, and how they co-evolve with our CEX quantiles. Appendix [E](#) presents residual diagnostics for our estimated model. At this point, a reader not interested in those details should just know that we used DMD to estimate the parameters of the VAR [\(12\)](#) that drives the additive functional for aggregate income Y_t . The next subsection tells what that VAR teaches us about micro-macro interactions in our CEX cross-sections.

4.2 Micro-Macro Interactions

To investigate the influences of aggregate income growth $Y_t - Y_{t-1} - \nu$ on inequality dynamics, we compute population least squares regressions of innovations to various quantile differences on innovations to $Y_t - Y_{t-1} - \nu$. We can define a vector $\mathbf{b}$ so that a quantile

difference can be written as $z_{t+1} = \mathbf{b} \mathbf{y}_{t+1}$.¹⁹ Different choices of $\mathbf{b}$ correspond to different quantile differences. Let $\mathbf{a}_{z,t+1} := \mathbf{b} \mathbf{a}_{t+1}$ and $\mathbf{a}_{1,t+1} := e_1 \mathbf{a}_{t+1}$. We then project $\mathbf{a}_{z,t+1}$ on $\mathbf{a}_{1,t+1}$:

$$\mathbf{a}_{z,t+1} = \vartheta \mathbf{a}_{1,t+1} + \xi_{t+1} \quad (14)$$

The variance of $\mathbf{a}_{z,t+1}$ is $\mathbf{b} \mathbf{\Omega} \mathbf{b}^\top$ and the covariance $\text{Cov}(\mathbf{a}_{z,t+1}, \mathbf{a}_{1,t+1}) = \mathbf{b} \mathbf{\Omega} e_1^\top$, so the least squares regression coefficient is

$$\vartheta := \frac{\text{Cov}(\mathbf{a}_{z,t+1}, \mathbf{a}_{1,t+1})}{\text{Var}(\mathbf{a}_{1,t+1})} = \frac{\mathbf{b} \mathbf{\Omega} e_1^\top}{e_1 \mathbf{\Omega} e_1^\top}. \quad (15)$$

Hence, the fraction of variance of the innovation $\mathbf{a}_{z,t+1}$ in the quantile difference attributable to the innovation to aggregate income growth $\mathbf{a}_{1,t+1}$ is

$$\frac{(\mathbf{b} \mathbf{\Omega} e_1^\top)^2}{(\mathbf{b} \mathbf{\Omega} \mathbf{b}^\top)(e_1 \mathbf{\Omega} e_1^\top)} = \frac{\vartheta^2 (e_1 \mathbf{\Omega} e_1^\top)}{\mathbf{b} \mathbf{\Omega} \mathbf{b}^\top} \quad (16)$$

The three panels of Figure 6 plot variance ratio (16) for several $p_{\text{high}} - p_{\text{low}}$ quantile differences of private income, post-tax income, and consumption. To show how the variance ratio varies with the choice of percentiles, the panels report quantile differences parameterized by $\mathbf{b} \mathbf{y}_{t+1}$ ranging from 90-10 percentile gaps to 81-17 percentile gaps. For private income, fractions (16) range between 15% and 31%, with the highest values for the mid-range percentile differences (82-18 percentile). For post-tax income and consumption cross-sections, variance ratios (16) are smaller. Figure 6 shows that aggregate income is a larger source of variations in inequality in private incomes than of inequalities in either post-tax incomes or consumption. Figure 6 thus indicates the presence of pervasive insurance against aggregate business cycle risk in the CEX consumption cross-sections from 1990-2023.²⁰

¹⁹For instance, with $\mathbf{y}_t$ stacked as in (5), a 90–10 private-income gap corresponds to $\mathbf{b} = [0 \quad (e_{90}^{(m)} - e_{10}^{(m)})^\top \quad \mathbf{0}_{1 \times M} \quad \mathbf{0}_{1 \times M}]$, where $e_p^{(m)} \in \mathbb{R}^M$ is the canonical basis vector for percentile p in the private-income block.

²⁰Gramm et al. (2024, Fig. 2.1) present striking graphical evidence of substantial US redistribution of income via taxes and transfers.

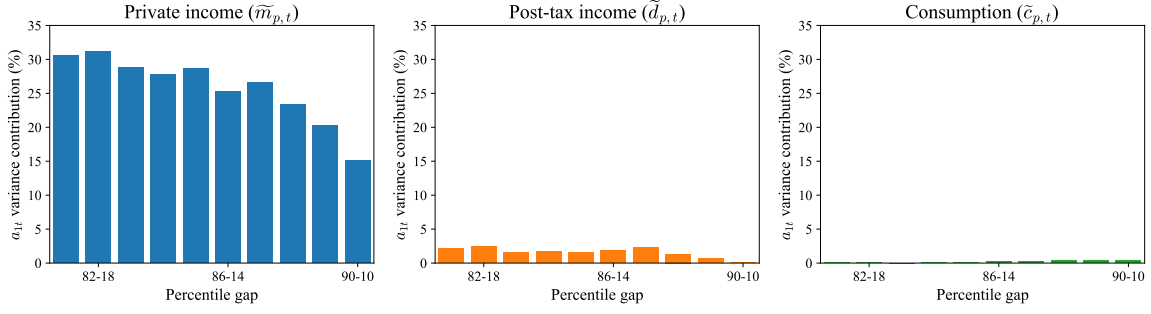


Figure 6: Contributions of aggregate income to inequality

5 Welfare Comparisons

The gains from removing *all* existing variability in aggregate consumption . . . are surely well below 1 percent of national income. Policies that deal with the very real problems of society’s less fortunate – wealth redistribution and social insurance – can be designed in total ignorance of the nature of business-cycle dynamics.

— Lucas (1987, p. 105)

This section uses our statistical model of the co-evolution of quantile-specific private income, private income net of taxes and transfers, and consumption to extend welfare calculations of Lucas (1987, 2003). These calculations confirm and broaden a claim of Lucas (1987, p. 105) that redistribution and insurance provided by the US tax-and-transfer system have raised individual consumers’ discounted utilities substantially more than would further reductions of aggregate fluctuations around trend growth rates. After Section 5.1 reviews Lucas’s welfare calculations and the statistical model with its i.i.d. risk in per-capita consumption that it assumes, Section 5.2 studies quantile-by-quantile welfare benefits from removing exposures to risks in consumption growth rates under our statistical model’s representation of serially correlated risks in quantile-specific consumption growth rates. Section 5.3 then studies cross-quantile welfare consequences of alterations in the trend rate of growth. Section 5.4 computes additional value functions and uses them to study cross-quantile welfare consequences of participating in the US tax-and-transfer system. Section 5.5 decomposes welfare costs of consumption risk into contributions from permanent and transitory shocks.

5.1 Lucas's Random Walk Specification

To set the stage for our extensions of their welfare calculations, we begin by telling how [Lucas \(1987, Sec. III\)](#), [Lucas \(2003\)](#), and [Tallarini \(2000\)](#) calculated welfare consequences of removing a representative consumer's exposure to i.i.d. risk in US post WWII consumption growth per capita.²¹ They let c_t be the log of consumption per capita C_t and assume that a time-0 representative agent orders consumption streams according to the discounted expected value

$$v(c_0) = (1 - \beta) \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t c_t, \quad (17)$$

where $\beta \in (0, 1)$ is a discount factor, and c_t evolves according to

$$c_{t+1} - c_t = \mu + \sigma_\epsilon \epsilon_{t+1}, \quad \epsilon_{t+1} \sim \mathcal{N}(0, 1). \quad (18)$$

Under this specification, the conditional expectation of $C_{t+1} = \exp(c_{t+1})$ is

$$\mathbb{E}_t C_{t+1} = \exp\left(\mu + \frac{1}{2}\sigma_\epsilon^2\right) C_t, \quad (19)$$

and the expected discounted value (17) has the form

$$v(c_0) = c_0 + \frac{\beta}{(1 - \beta)} \mu. \quad (20)$$

The value of a risk-free path that starts at $\bar{c}_0$ at $t = 0$ and has the same conditional means $\mathbb{E}_0 C_{t+j}, j \geq 0$, as the risky sequence generated by (18) is

$$v(\bar{c}_0) = \bar{c}_0 + \frac{\beta}{(1 - \beta)} \left(\mu + \frac{1}{2}\sigma_\epsilon^2\right). \quad (21)$$

Setting $v(c_0) = v(\bar{c}_0)$, we derive the compensating difference

$$c_0 - \bar{c}_0 = \frac{\beta \sigma_\epsilon^2}{2(1 - \beta)}. \quad (22)$$

²¹[Obstfeld \(1994\)](#), [Dolmas \(1998\)](#), [Hansen et al. \(1999\)](#), [Alvarez and Jermann \(2004\)](#), and [Barillas et al. \(2009\)](#) studied extensions including some that were based on discrete-time versions of risk-sensitive value functions and other recursive utility functionals.

Hence, equation (22) is a “compensating difference” that equates (20) to (21). The gap $c_0 - \bar{c}_0$ in equation (22) is a percentage of consumption that a representative consumer would be willing to sacrifice if he could instead live in a parallel economy in which σ_ϵ has been set to zero as a result of ideal countercyclical monetary and fiscal policies. Lucas (1987, Sec. III) and Lucas (2003) interpreted gap $c_0 - \bar{c}_0$ as an upper bound on welfare gains that could be gathered by post WWII countercyclical US monetary and fiscal policies.

5.2 Welfare Costs of Martingale and Stationary Shocks

The logarithmic random walk-with-drift specification (18) on which Lucas’s formula (22) is based exposes a representative consumer to i.i.d. risk in consumption growth. Our specification

$$c_{p,t} = \underbrace{t\nu}_{\text{Trend}} + \underbrace{\sum_{j=1}^t \mathbf{H} \mathbf{a}_j}_{\text{Martingale}} \underbrace{-(\mathbf{g} - e_p^{(c)}) \mathbf{y}_t}_{\text{Stationary}} + \underbrace{(\mathbf{g} \mathbf{y}_0 + Y_0 + \nu_p^{(c)})}_{\text{Constant}} \quad (23)$$

exposes a consumer in the p th quantile to i.i.d. risk in consumption growth through $\mathbf{a}_{t+1}$ and to serially correlated risk in consumption growth through the martingale $\sum_{j=1}^t \mathbf{H} \mathbf{a}_j$ and the serially correlated state variables $\mathbf{y}_t$.²²

We shall calculate quantile-specific equalizing differences for hypothetical consumers who are also exposed to quantile-specific serially correlated risks in their consumption growth rates. We compute percentages of current consumption that someone stuck forever in the p th consumption quantile would be willing to sacrifice in order to eliminate both the stationary *and* the martingale random components on the right hand side of representation (23).²³

We model a household permanently in the p th consumption quantile who values consumption streams at time t according to²⁴

$$v_p(\zeta_t, c_{p,t}) = (1 - \beta) \mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j c_{p,t+j} \right], \quad \beta \in (0, 1). \quad (24)$$

²²Bansal and Yaron (2004) imposed cross-equation restrictions from consumption Euler equations for US asset prices to infer the presence of such serially correlated risks in US aggregate consumption. Hansen et al. (2008) and Hansen and Sargent (2010) also studied long-run risks but intentionally imposed those cross-equation restrictions only at a later stage in their analysis.

²³These calculations can be extended to include a Markov process of transitions across quantiles in future research.

²⁴In our exercises, we set $\beta = 0.99$ for quarterly data.

To link this to (6), define the augmented state $\zeta_t = \begin{bmatrix} 1 \\ \mathbf{y}_t \end{bmatrix}$ and the transition $\mathbf{A} = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}$. Let $\mathbf{a}_{t+1} = \mathbf{J}\epsilon_{t+1}$ where $\epsilon_{t+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{\Omega} = \mathbf{J}\mathbf{J}^\top$. Choose $\mathbf{C} = \begin{bmatrix} \mathbf{0} \\ \mathbf{J} \end{bmatrix}$ so that the lower block of $\zeta_{t+1} = \mathbf{A}\zeta_t + \mathbf{C}\epsilon_{t+1}$ reproduces $\mathbf{y}_{t+1} = \mathbf{B}\mathbf{y}_t + \mathbf{J}\epsilon_{t+1}$ in (6).

Define the loadings

$$\mathbf{D}_p^{(c)} := \begin{bmatrix} \nu & e_1 \mathbf{B} + e_p^{(c)}(\mathbf{B} - \mathbf{I}) \end{bmatrix}, \quad (25)$$

$$\mathbf{F}_p^{(c)} := (e_1 + e_p^{(c)})\mathbf{J}. \quad (26)$$

These expressions follow from differencing (23), substituting $\mathbf{y}_{t+1} = \mathbf{B}\mathbf{y}_t + \mathbf{a}_{t+1}$ with $\mathbf{a}_{t+1} = \mathbf{J}\epsilon_{t+1}$, and collecting terms in ζ_t and ϵ_{t+1} . The state-space system for consumption growth for percentile p is

$$\zeta_{t+1} = \mathbf{A}\zeta_t + \mathbf{C}\epsilon_{t+1}, \quad (27)$$

$$c_{p,t+1} - c_{p,t} = \mathbf{D}_p^{(c)}\zeta_t + \mathbf{F}_p^{(c)}\epsilon_{t+1}. \quad (28)$$

In this representation, $\mathbf{F}_p^{(c)}\epsilon_{t+1}$ captures exposure to i.i.d. risk in quantile- p consumption growth, while $\mathbf{D}_p^{(c)}\zeta_t$ captures exposure to persistent risk.

We seek a value function $v_p(\zeta, c_{p,t})$ that satisfies the recursion

$$v_p(\zeta, c_p) = (1 - \beta)c_p + \beta \mathbb{E}[v_p(\zeta', c'_p)]$$

where ζ, c_p are current period's state and consumption, and ζ', c'_p are next period's state and consumption satisfying (27)–(28). We use this simplified notation for recursive representations in the remainder of this section. The following proposition uses the logic of Hansen et al. (2008) and Hansen and Sargent (2010) to compute value functions.

Proposition 2. *The value function for a consumer permanently in the p th consumption quantile is linear in the state ζ and current log consumption c_p :*

$$v_p(\zeta, c_p) = c_p + \boldsymbol{\lambda}_p^{(c)\top} \zeta, \quad (29)$$

where

$$\boldsymbol{\lambda}_p^{(c)\top} = \beta \mathbf{D}_p^{(c)} (\mathbf{I} - \beta \mathbf{A})^{-1}. \quad (30)$$

Proof. See Appendix B.3 □

To compare the quantile- p value function (29) to one associated with a risk-free certainty equivalent consumption path (in the spirit of Lucas (1987, 2003); Tallarini (2000)), we iterate (27)–(28) and derive by induction, for $j \geq 1$,

$$c_{p,t+j} = c_{p,t} + \boldsymbol{\psi}_{p,j}^{(c)\top} \boldsymbol{\zeta}_t + \sum_{i=0}^{j-1} \mathbf{J}_{p,i}^{(c)} \boldsymbol{\epsilon}_{t+j-i}, \quad (31)$$

where

$$\boldsymbol{\psi}_{p,j}^{(c)\top} = \sum_{m=0}^{j-1} \mathbf{D}_p^{(c)} \mathbf{A}^m, \quad (32)$$

$$\mathbf{J}_{p,i}^{(c)} = \mathbf{F}_p^{(c)} + \sum_{\ell=0}^{i-1} \mathbf{D}_p^{(c)} \mathbf{A}^\ell \mathbf{C}, \quad i \geq 0. \quad (33)$$

The coefficient $\mathbf{J}_{p,i}^{(c)}$ in (33) represents the cumulative loading of $c_{p,t+j}$ on the shock $\boldsymbol{\epsilon}_{t+j-i}$ that arrives i steps earlier: a direct effect via $\mathbf{F}_p^{(c)}$ plus indirect effects where the shock first moves the state (through $\mathbf{C}$), the state propagates (through $\mathbf{A}^\ell$), and then affects consumption growth (through $\mathbf{D}_p^{(c)}$).

To construct a counterpart to Lucas's certainty equivalent value function formula (21) under Lucas's statistical model for consumption, we define the certainty equivalent value function

$$w_p(\boldsymbol{\zeta}, c_p^{CE}) = c_p^{CE} + \boldsymbol{\lambda}_p^{(c)\top} \boldsymbol{\zeta} + \alpha_p^{(c)}, \quad (34)$$

where c_p^{CE} is a certainty equivalent (log) consumption at time t for a consumer in the p th quantile, and $\alpha_p^{(c)} := c_p - c_p^{CE}$ is a compensating differential that adjusts for the risk in the consumption growth process defined by

$$\alpha_p^{(c)} = \frac{\beta}{2} \sum_{i=0}^{\infty} \beta^i \mathbf{J}_{p,i}^{(c)} \mathbf{J}_{p,i}^{(c)\top}, \quad (35)$$

where $\mathbf{J}_{p,i}^{(c)}$ is defined in (33). Equation (35) follows by equating the discounted value of the risky path to that of a risk-free path with the same conditional expectations for levels $C_{p,t+j} = \exp(c_{p,t+j})$, using the log-normal moment formula applied to (31) in the spirit of Lucas (1987).

The compensation $\alpha_p^{(c)}$ depends on both i.i.d. risk (through $\mathbf{F}_p^{(c)}$ in (26)) and persistent risk (through the state loadings $\mathbf{D}_p^{(c)}$ in (25)). This compensation differs across quantiles but is constant across time, a pattern built into our specification of the additive process (8) for Y_t .

To measure the benefits from redistribution and insurance, we temporarily pretend that p th quantile households must consume either total private income $m_{p,t}$ or income net of taxes and transfers $d_{p,t}$ instead of the $c_{p,t}$ that they actually consume. Comparing value functions associated with consuming private income provides us with measures of welfare gains to a p th quantile consumer from participating in the US tax-and-transfer system. Comparing value functions for consuming income net of transfers and for consumption allows us to measure benefits that are presumably achieved through management of personal savings.²⁵

The three panels of Figure 7 plot quantile-specific compensations for eliminating all risks from growth rates in private income (left), post-tax income (middle), and consumption (right). In each panel, the colored region shows the i.i.d. component $\frac{\beta}{2} \mathbf{F}_p^{(u)} \mathbf{F}_p^{(u)\top}$ and the gray region shows the persistent component; together they sum to total compensation $\alpha_p^{(u)}$. All three panels display a U-shape across quantiles, with compensations highest at the tails and lowest in the middle of the distribution. For consumption, $\alpha_p^{(c)}$ reaches its minimum around the 50th percentile, while for private income $\alpha_p^{(m)}$ reaches its minimum around the low-50s. This pattern is largely driven by loadings $(e_1 + e_p^{(c)})\mathbf{J}$ in formula (26) for $\mathbf{F}_p^{(c)} \mathbf{F}_p^{(c)\top}$: low-quantile households face high idiosyncratic volatility on their labor income. While high-quantile households often have stable labor income, they face greater exposure to market risks through business income and capital gains. Figure 8 focuses specifically on the i.i.d. components of the compensations in Figure 7. The i.i.d. component is a more important contributor at the tails than for the middle-quantile households. Agents in the 50th percentile of the consumption distribution would be willing to pay about 2.3 percent of consumption to remove all risk in

²⁵Our joint modelling of the three CEX data series make it possible for us to entertain mental experiments like this. Lucas (1987, Sec. III), Lucas (2003), and Tallarini (2000) based their calculations on calibrated versions of the exogenous consumption endowment, representative-agent economy of Lucas (1978).

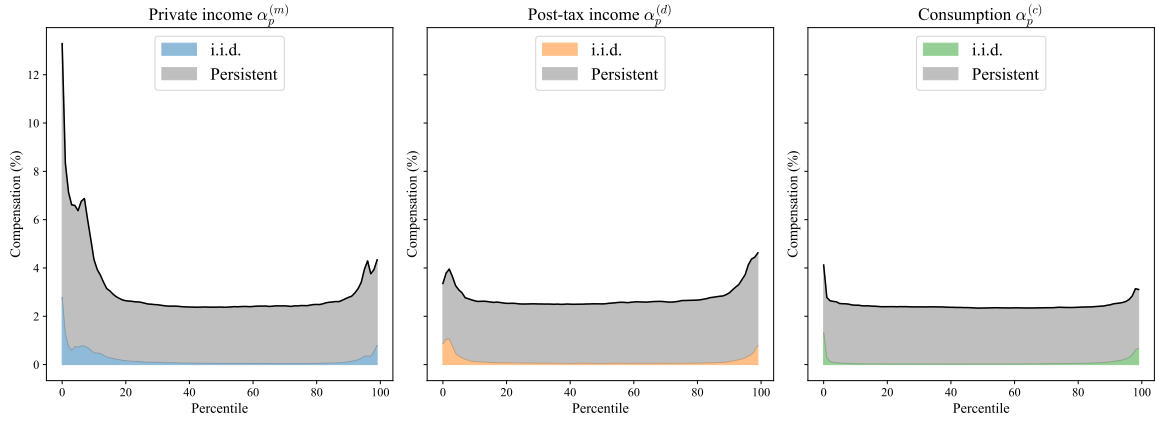


Figure 7: Compensations for eliminating risks in income or consumption growth

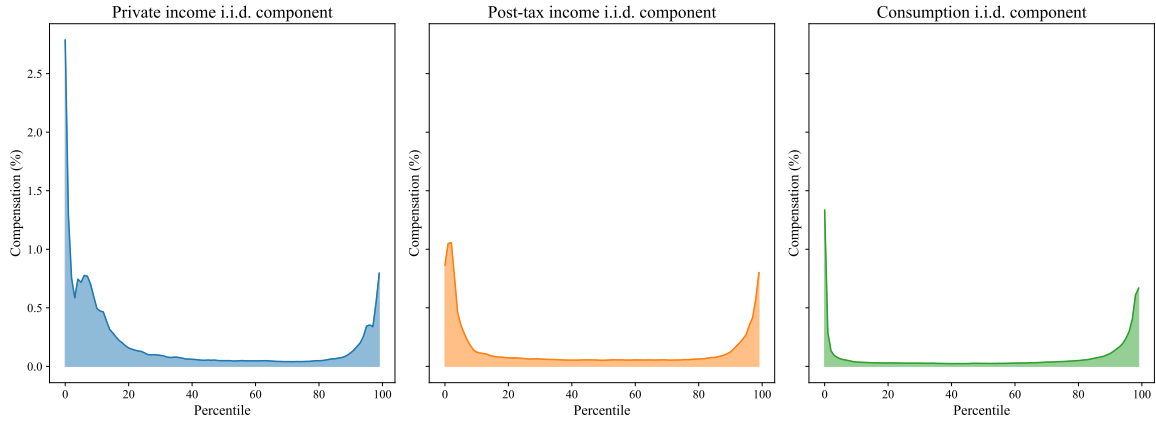


Figure 8: Compensations for eliminating i.i.d. risks in income or consumption growth

consumption growth. This is approximately 45 times larger than the 0.05% of consumption for Lucas's (2003) representative consumer. Our estimate for i.i.d. risk is about 0.03% of consumption, which is close to Lucas's estimate. The remaining part in our estimate comes from exposure to serially correlated risk in consumption growth.²⁶

For all quantiles, proportional consumption compensations are smaller than proportional compensations for private and post-tax incomes, while these proportional compensations are similar for all quantiles. These patterns indicate that decrements in values due to exposures to consumption growth risks are lower than for exposures to income risks.

²⁶Bansal and Yaron (2004), Hansen et al. (2008), and Hansen and Sargent (2010) explore implications for market prices of risk of exposing representative consumers who dislike it to *very* persistent risk in consumption growth. Bansal and Yaron call it "long-run risk".

5.3 Welfare consequences of growth

In the spirit of [Lucas \(1987, 2003\)](#) and [Tallarini \(2000\)](#), we now calculate reductions in consumption levels that households would be willing to accept in exchange for a once-and-for-all increase in the trend growth rate ν . The discounted expected utility of someone who is permanently in the p th consumption quantile of an economy with aggregate income trend growth $\bar{\nu} > \nu$ is

$$v_p(\zeta, \bar{c}_p) = \bar{c}_p + (\bar{\lambda}_p^{(c)})^\top \zeta, \quad (36)$$

where $(\bar{\lambda}_p^{(c)})^\top = \beta \bar{\mathbf{D}}_p^{(c)} (\mathbf{I} - \beta \mathbf{A})^{-1}$ and $\bar{\mathbf{D}}_p^{(c)} = \begin{bmatrix} \bar{\nu} & e_1 \mathbf{B} + e_p^{(c)} (\mathbf{B} - \mathbf{I}) \end{bmatrix}$. A proportional decrease in the consumption that leaves a p th quantile consumer indifferent between ν and $\bar{\nu}$ is

$$\begin{aligned} c_p - \bar{c}_p &= (\bar{\lambda}_p^{(c)\top} - \lambda_p^{(c)\top}) \zeta \\ &= \beta (\bar{\mathbf{D}}_p^{(c)} - \mathbf{D}_p^{(c)}) (\mathbf{I} - \beta \mathbf{A})^{-1} \zeta. \end{aligned} \quad (37)$$

Using definitions of $\mathbf{D}_p^{(c)}$, $\bar{\mathbf{D}}_p^{(c)}$ and ζ , we can simplify this to

$$c_p - \bar{c}_p = \beta (\bar{\nu} - \nu) \phi_1(\zeta), \quad (38)$$

where $\phi_1(\zeta)$ is the first element of the vector $(\mathbf{I} - \beta \mathbf{A})^{-1} \zeta$. The compensating difference is constant across time and quantiles, another feature that our additive process (8) builds in.

Formula (38) indicates that households in all quantiles would be willing to sacrifice roughly 25 percent of current consumption to increase aggregate trend growth ν by 1 percentage point, i.e., from 1.3% to 2.3%. This reaffirms [Lucas's](#) (p. 1, 2003) conclusion that “*the potential for welfare gains from better long-run, supply-side policies exceeds by far the potential from further improvements in short-run demand management.*”

To compare the value of growth with the value of eliminating consumption risk, we transform (38) to compute the increment in trend growth that would leave a p th quantile consumer as well off as if all consumption risk were removed:

$$\bar{\nu} - \nu = \frac{(c_p - \bar{c}_p)}{\beta \phi_1(\zeta)} = \frac{\alpha_p^{(c)}}{\beta \phi_1(\zeta)} \quad (39)$$

We also compute counterpart comparisons for households constrained to consume total private income and for households consuming income net of transfers and taxes. To decompose the sources of these welfare differences, we write

$$v_p(\zeta, c_p) - v_p(\zeta, m_p) = (c_p - m_p) + (\lambda_p^{(c)\top} - \lambda_p^{(m)\top})\zeta \quad (40)$$

where the first term measures the increment in value attributable to redistribution, while the second term captures insurance against persistent risk. More explicitly, the welfare difference at time t and state ζ_t is

$$\Delta v_{p,t} := v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t}) = (c_{p,t} - m_{p,t}) + (\lambda_p^{(c)\top} - \lambda_p^{(m)\top})\zeta_t.$$

Applying the same linear mapping as in (38)–(39), we convert this welfare difference into a trend growth increment:

$$\tilde{\nu} - \nu = \frac{\Delta v_{p,t}}{\beta \phi_1(\zeta_t)}, \quad \phi_1(\zeta_t) := e_1^\top (\mathbf{I} - \beta \mathbf{A})^{-1} \zeta_t.$$

Table 1 reports these compensations for eliminating all risks in consumption growth by setting $\bar{c}_p = c_p^{CE}$, and the value of insurance for the 10th, 50th and 90th percentiles.²⁷ For all quantiles, the utility benefits of eliminating all risks to consumption growth are equivalent to just a 0.09–0.10 percentage-point increase in annual trend growth (see the first two columns of Table 1). In contrast, for a 10th percentile household, access to the US tax, transfer, and financial system is welfare-equivalent to an increase in annual trend growth of aggregate income by about 4.2 percentage points, whereas the corresponding increments are about –0.6 and –1.3 percentage points for the 50th and 90th percentiles, respectively (rightmost columns of the table).

5.4 Welfare consequences of US tax-and-transfer system

We can interpret $v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t})$ as measuring welfare benefits for a household in the p th quantile from managing personal savings and from participating in the US tax and transfer

²⁷For “eliminating all risk” we use (39) with $\alpha_p^{(c)}$. For “insurance” we map the value difference $\Delta v_{p,t} = v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t})$ into a growth increment via $\tilde{\nu} - \nu = \Delta v_{p,t} / (\beta \phi_1(\zeta_t))$.

Quantile	Value of eliminating all risk to consumption growth		Value of social and private insurance	
	$(c_p - c_p^{CE})$ %	$\bar{\nu} - \nu$ %, annual.	$v_p(\zeta, c_p) - v_p(\zeta, m_p)$ $\times 100$	$\tilde{\nu} - \nu$ %, annual.
$p = 10$	2.46	0.099	103.03	4.16
$p = 50$	2.34	0.095	-15.30	-0.62
$p = 90$	2.47	0.100	-32.07	-1.30

Table 1: Compensating differences for eliminating risks to consumption growth. Value-function differences in columns 3–4 are evaluated at end-of-sample state ζ_T .

system. Table 2 shows the mean, minimum, and maximum of $v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t})$ for $p = 10, 50, 90$ over quarterly CEX waves from 1990 to 2023. For the 10th percentile, the mean is about 0.94 and the values range from about 0.86 to 1.04, which are orders of magnitude larger than the gains from removing exposure to growth rate risks shown in Figure 7.²⁸ For the 90th percentile the mean is about -0.31 , and the value differences are negative throughout the sample. The negative numbers for 90th percentile households indicate that for them, participation in the US tax-and-transfer system is harmful and consequential, dwarfing the preceding hypothetical benefits from removing exposure to growth rate risks.

Quantile	$v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t})$		
	Mean	Min	Max
$p = 10$	0.9363	0.8637	1.0401
$p = 50$	-0.1580	-0.1619	-0.1526
$p = 90$	-0.3084	-0.3255	-0.2922

Table 2: Mean, minimum, and maximum of $v_p(\zeta_t, c_{p,t}) - v_p(\zeta_t, m_{p,t})$ from 1990 to 2023, for $p = 10, 50, 90$.

5.5 Welfare Costs of Permanent and Transitory Shocks

In Section 5.2, we computed the compensating differences $\alpha_p^{(c)}$ for eliminating all risk in consumption growth for households in each consumption quantile p . In this section, we study how much of these compensations are attributable to the permanent (martingale) component and to the transitory (stationary) component of consumption.

²⁸Footnote 16 briefly points toward a way to interpret these outcomes heuristically with a general equilibrium model with extensive efficient redistribution and insurance.

5.5.1 Welfare under Single-Component Processes

Because (23) decomposes quantile- p consumption levels into a martingale term $\sum_{j=1}^t \mathbf{H} \mathbf{a}_j$ and a stationary term $-(\mathbf{g} - e_p^{(c)}) \mathbf{y}_t$, we can construct consumption processes in which either the stationary term or the martingale term is set to zero.²⁹

Consider first a martingale-only consumption process defined by

$$c_{p,t}^{(M)} := t\nu + \sum_{j=1}^t \mathbf{H} \mathbf{a}_j. \quad (41)$$

Here, we use $\cdot^{(M)}$ to denote objects associated with the martingale-only process. Then consumption growth is given by

$$c_{p,t+1}^{(M)} - c_{p,t}^{(M)} = \nu + \mathbf{H} \mathbf{a}_{t+1} = \nu + (\mathbf{H}\mathbf{J})\epsilon_{t+1}. \quad (42)$$

In the state-space representation (27)–(28), we can keep $\mathbf{A}$ and $\mathbf{C}$ unchanged and set

$$\mathbf{D}^{(M)} := \begin{bmatrix} \nu & \mathbf{0} \end{bmatrix}, \quad \mathbf{F}^{(M)} := \mathbf{H}\mathbf{J}. \quad (43)$$

Since $\mathbf{D}^{(M)} \mathbf{A}^\ell \mathbf{C} = \mathbf{0}$ for all $\ell \geq 0$, (33) implies $\mathbf{J}_i^{(M)} = \mathbf{F}^{(M)}$ for all $i \geq 0$ and therefore the compensation in (35) reduces to

$$\alpha^{(M)} = \frac{\beta}{2(1-\beta)} \mathbf{F}^{(M)} (\mathbf{F}^{(M)})^\top = \frac{\beta}{2(1-\beta)} \mathbf{H} \mathbf{\Omega} \mathbf{H}^\top. \quad (44)$$

Compensation $\alpha^{(M)}$ does not depend on p because the martingale component is common across quantiles in our specification.

Consider next a stationary-only consumption process defined by

$$c_{p,t}^{(S)} := t\nu - (\mathbf{g} - e_p^{(c)}) \mathbf{y}_t. \quad (45)$$

Here, we use $\cdot^{(S)}$ to denote objects associated with the stationary-component-only process.

Differencing and substituting $\mathbf{y}_{t+1} - \mathbf{y}_t = (\mathbf{B} - \mathbf{I}) \mathbf{y}_t + \mathbf{a}_{t+1}$ and setting $\mathbf{D}_p^{(S)} := \mathbf{D}_p^{(c)}$ and

²⁹Constants in (23) do not affect consumption growth or compensating differences, so we omit constant terms below.

$\mathbf{F}_p^{(S)} := -(\mathbf{g} - e_p^{(c)})\mathbf{J}$ give

$$\begin{aligned} c_{p,t+1}^{(S)} - c_{p,t}^{(S)} &= \nu - (\mathbf{g} - e_p^{(c)})((\mathbf{B} - \mathbf{I})\mathbf{y}_t + \mathbf{a}_{t+1}) \\ &= \nu + (e_1 \mathbf{B} + e_p^{(c)}(\mathbf{B} - \mathbf{I}))\mathbf{y}_t - (\mathbf{g} - e_p^{(c)})\mathbf{a}_{t+1} \\ &= \mathbf{D}_p^{(c)}\boldsymbol{\zeta}_t + \mathbf{F}_p^{(S)}\boldsymbol{\epsilon}_{t+1}, \end{aligned} \quad (46)$$

where we used $\mathbf{g}(\mathbf{I} - \mathbf{B}) = e_1 \mathbf{B}$ from (10) in the second equality. Thus, setting the martingale term in (23) to zero leaves the state loading unchanged but modifies the innovation loading in (28). The corresponding shock-loading sequence is

$$\mathbf{J}_{p,i}^{(S)} := \mathbf{F}_p^{(S)} + \sum_{\ell=0}^{i-1} \mathbf{D}_p^{(c)} \mathbf{A}^\ell \mathbf{C}, \quad i \geq 0, \quad (47)$$

and the stationary-component-only compensation is

$$\alpha_p^{(S)} := \frac{\beta}{2} \sum_{i=0}^{\infty} \beta^i \mathbf{J}_{p,i}^{(S)} \mathbf{J}_{p,i}^{(S)\top}. \quad (48)$$

Because the same shock $\mathbf{a}_{t+1}$ appears in the martingale term and drives $\mathbf{y}_t$ in the stationary term of (23), in general $\alpha_p^{(c)}$ is not equal to $\alpha^{(M)} + \alpha_p^{(S)}$. Table 3 reports welfare costs under these single-component processes for $p = 10, 50, 90$. The permanent-component-only compensation $\alpha^{(M)}$ is 2.58% across all quantiles, while the transitory-component-only compensations $\alpha_p^{(S)}$ range from 0.05% to 0.17%. Note that the permanent-component-only compensation $\alpha^{(M)}$ exceeds the total compensation $\alpha_p^{(c)}$. This occurs because the cumulative loadings partially offset each other. Since the martingale-component-only loading is $\mathbf{J}_i^{(M)} = \mathbf{F}^{(M)}$ for all i (because $\mathbf{D}^{(M)} \mathbf{A}^\ell \mathbf{C} = \mathbf{0}$), while the original cumulative loading $\mathbf{J}_{p,i}^{(c)}$ satisfies $\mathbf{J}_{p,i}^{(c)} = \mathbf{F}^{(M)} + \mathbf{J}_{p,i}^{(S)}$, we have

$$\|\mathbf{F}^{(M)}\|^2 + \|\mathbf{J}_{p,i}^{(S)}\|^2 - \|\mathbf{J}_{p,i}^{(c)}\|^2 = -2\langle \mathbf{F}^{(M)}, \mathbf{J}_{p,i}^{(S)} \rangle.$$

When this cross-term $\langle \mathbf{F}^{(M)}, \mathbf{J}_{p,i}^{(S)} \rangle$ is positive across horizons i , the full process has lower welfare costs than the sum of the single-component processes.

Quantile	$\alpha_p^{(c)}$ %	$\alpha^{(M)}$ %	$\alpha_p^{(S)}$ %	$(\alpha^{(M)} + \alpha_p^{(S)} - \alpha_p^{(c)})$ %
$p = 10$	2.46	2.58	0.05	0.17
$p = 50$	2.34	2.58	0.13	0.36
$p = 90$	2.47	2.58	0.17	0.27

Table 3: Welfare costs under single-component processes. $\alpha^{(M)}$ is the compensation when only the permanent (martingale) component is active; $\alpha_p^{(S)}$ is the compensation when only the transitory (stationary) component is active.

5.5.2 Decomposition via Projection onto the Martingale Component

In Section 5.5.1, we shut down either the permanent or transitory component entirely. Since the error $\mathbf{a}_t$ appears in both stationary and martingale terms, they do not add up to the total compensation of removing all risks $\alpha_p^{(c)}$. In this section, we derive an alternative decomposition that projects the cumulative shock loadings $\mathbf{J}_{p,i}^{(c)}$ into a component aligned with the innovation associated with the martingale component of (23):

$$\mathbf{a}_{H,t+1} := \mathbf{H} \mathbf{a}_{t+1} = (\mathbf{H}\mathbf{J})\boldsymbol{\epsilon}_{t+1},$$

and computes its orthogonal complement.

Recall from (35) that the welfare compensation depends on $\mathbf{J}_{p,i}^{(c)}\mathbf{J}_{p,i}^{(c)\top}$, the squared norm of the cumulative loading on the i.i.d. shocks $\boldsymbol{\epsilon}_{t+1}$. Define the unit vector

$$\mathbf{b} := \frac{\mathbf{H}\mathbf{J}}{\|\mathbf{H}\mathbf{J}\|} = \frac{\mathbf{H}\mathbf{J}}{\sigma_H}, \quad \sigma_H := \sqrt{\mathbf{H}\boldsymbol{\Omega}\mathbf{H}^\top}. \quad (49)$$

This vector defines the direction that drives the martingale component of (23). Using (49), the martingale component of consumption growth $\mathbf{a}_{H,t+1}$ can be written as $\mathbf{a}_{H,t+1} = \sigma_H(\mathbf{b} \cdot \boldsymbol{\epsilon}_{t+1})$.

We can decompose vector $\mathbf{J}_{p,i}^{(c)}$ into distinct components that are parallel and orthogonal to $\mathbf{b}$:

$$\|\mathbf{J}_{p,i}^{(c)}\|^2 = \mathbf{J}_{p,i}^{(c)}\mathbf{J}_{p,i}^{(c)\top} = \underbrace{(\mathbf{J}_{p,i}^{(c)} \cdot \mathbf{b})^2}_{\text{parallel to } \mathbf{b}} + \underbrace{\|\mathbf{J}_{p,i}^{(c)} - (\mathbf{J}_{p,i}^{(c)} \cdot \mathbf{b})\mathbf{b}\|^2}_{\text{orthogonal to } \mathbf{b}}. \quad (50)$$

The first term projects the total risk onto the direction of the martingale component, and the second term captures risk orthogonal to it.

Since $\boldsymbol{\epsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, projections onto orthogonal directions are independent. Substituting

(50) into (35) yields the decomposition

$$\alpha_p^{(c)} = \alpha_p^{\text{mart}} + \alpha_p^{\text{orth}}, \quad (51)$$

where

$$\alpha_p^{\text{mart}} := \frac{\beta}{2} \sum_{i=0}^{\infty} \beta^i (\mathbf{J}_{p,i}^{(c)} \cdot \mathbf{b})^2, \quad (52)$$

$$\alpha_p^{\text{orth}} := \frac{\beta}{2} \sum_{i=0}^{\infty} \beta^i [\|\mathbf{J}_{p,i}^{(c)}\|^2 - (\mathbf{J}_{p,i}^{(c)} \cdot \mathbf{b})^2]. \quad (53)$$

The term α_p^{mart} captures welfare costs from $\mathbf{b} \cdot \epsilon_{t+1}$, the component of ϵ_{t+1} that drives the martingale, while α_p^{orth} captures costs from components of ϵ_{t+1} orthogonal to $\mathbf{b}$. The decomposition is exact because we are partitioning variance across orthogonal components of ϵ_{t+1} , not across different processes that share shocks. The two components are mutually orthogonal.³⁰

Table 4 reports this orthogonal decomposition of consumption compensating differences for $p = 10, 50, 90$. Unlike the result in Table 3, the contributions here sum exactly to the total because the decomposition partitions variance across orthogonal components of ϵ_{t+1} .

Quantile	$\alpha_p^{(c)}$ %	α_p^{mart} %	α_p^{orth} %	permanent share %
$p = 10$	2.46	2.41	0.04	98.3
$p = 50$	2.34	2.24	0.10	95.8
$p = 90$	2.47	2.32	0.15	94.0

Table 4: Decomposition of consumption compensating differences via orthogonal shock rotation. α_p^{mart} is the contribution from the permanent (martingale) innovation; α_p^{orth} is the contribution from innovations orthogonal to it.

Note that $\alpha_p^{\text{mart}} < \alpha^{(M)}$ for all p . This reflects the same observation we made in Table 3: $\alpha^{(M)}$ is computed using the martingale-only loadings $\mathbf{J}_i^{(M)} = \mathbf{F}^{(M)}$, whose projection onto $\mathbf{b}$ is $\mathbf{F}^{(M)} \cdot \mathbf{b} = \sigma_H$. In contrast, α_p^{mart} uses loadings $\mathbf{J}_{p,i}^{(c)}$, whose projection $\mathbf{J}_{p,i}^{(c)} \cdot \mathbf{b}$ is reduced by the stationary term's cumulative loading $\mathbf{J}_{p,i}^{(S)}$ when it has a negative component in the

³⁰Equivalently, one can construct an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ with first row $\mathbf{b}^\top$ and define rotated shocks $\tilde{\epsilon}_{t+1} = \mathbf{Q}\epsilon_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The first component $\tilde{\epsilon}_{1,t+1} = \mathbf{b} \cdot \epsilon_{t+1}$ is the standardized innovation in the direction that drives the martingale component. Since $\mathbf{Q}$ is orthogonal, $\|\mathbf{J}_{p,i}^{(c)}\|^2 = \|\mathbf{J}_{p,i}^{(c)} \mathbf{Q}^\top\|^2$, and the decomposition (52)–(53) partitions variance across the n independent components of $\tilde{\epsilon}_{t+1}$.

b direction. The permanent risk accounts for 94–98% of $\alpha_p^{(c)}$ at these quantiles, with the remaining 2–6% attributable to innovations orthogonal to it, which is qualitatively similar to the single-component analysis in Table 3.

6 Concluding Remarks

We have used additive functionals and dynamic mode decompositions to describe the coevolution of CEX cross-sections of private income, post-tax income, and consumption from 1990 to 2023. We decomposed a Chisini mean of private income and cross-section CEX quantiles of private earned incomes, incomes after taxes and transfers, and consumption levels into constant, deterministic trend, martingale, and stationary components. We use our statistical model to analyze how aggregate income has evolved along with those cross sections.

Our decomposition of aggregate income indicates that aggregate income was briefly above its deterministic trend before the 2008 financial crisis. After 2008, the martingale component fell suddenly and the stationary component fell gradually. By the end of the sample, the martingale component had recovered to about minus 10 percent and the stationary component was negative, dragging aggregate income below trend. There is substantial heterogeneity in stationary components of quantiles. For 10th percentile households, private income fell substantially after the financial crisis, stayed below deterministic trend and fell further during and after COVID-19. Nevertheless, for 10th percentile households, both post-tax income and consumption fell less and by 2023 had returned close to trend. For the 90th percentile, private income remained around the deterministic trend, as did post-tax income, but consumption remained below trend after the drop in 2008.

For private income inequality measures (e.g., 90-10 percentile differences), the fraction of variance attributable to aggregate income innovations ranges from around 15% to 31%. These fractions are smaller for post-tax income and consumption, indicating that redistribution and insurance arrangements have shaped the transmission of aggregate shocks to inequality.

Our extension of Lucas’s welfare calculations (Lucas, 1987, 2003) to heterogeneous consumers exposed to serially correlated consumption growth risks yields several findings. Consumers in the 50th percentile of the consumption distribution would sacrifice approximately

2.3% of current consumption to eliminate all consumption growth risk—approximately 45 times larger than Lucas’s representative agent estimate of 0.05%. But when we eliminate only i.i.d. risks, our estimates are similar to Lucas’s, indicating that persistent components of risks account for nearly all of the compensating differences. Even so, benefits of eliminating the consumption growth risk are small when compared to benefits from increasing trend growth. We use our additive functional model to compute that eliminating all consumption growth risk is welfare equivalent to just a 0.09–0.10 percentage point increase in annual trend growth.

Benefits from participating in the US tax-and-transfer system dwarf the benefits of eliminating consumption growth risks. For 10th percentile consumers, the value increment from the tax and transfer system is orders of magnitude larger than the welfare cost of consumption growth risk. For this group, the benefit is equivalent to roughly a 4.2 percentage point increase in annual trend growth. For 90th percentile consumers, the redistribution through the tax-and-transfer system imposes welfare costs that exceed the welfare costs of growth rate risk.

By projecting cumulative shock loadings onto the martingale innovation direction, we decompose consumption compensating differences into contributions from permanent and transitory risk: across quantiles, 94–98% of welfare costs stem from the permanent component, while innovations orthogonal to the permanent component contribute only 2–6%.

As for future research, the statistical framework that we have used to describe the joint dynamics of aggregate income and cross sections of incomes and consumption can serve as a good-fitting approximation that empirical implementations of structural HANK models should seek to match. It would also be interesting to extend our framework to allow households to move stochastically across consumption and income quantiles. In addition, we would like to do welfare calculations for households having preferences that express fears of model misspecification. Such households would be more averse to persistent risk in consumption growth than are the households modeled in this paper. We leave these and other useful extensions to later.

Appendices

A Data

In addition to the data processing steps described in Section 2, we make the following adjustments to the CEX data. In April 2013, the Bureau of Labor Statistics introduced a new methodology for estimating income tax liabilities and after-tax income in the Consumer Expenditure Survey, based on the integration of the TAXSIM model. This redesign generates a discrete level shift in the published post-tax income series beginning in 2013.³¹ To preserve comparability over time, we splice the post-tax income series at the 2013 break, constructing a consistent series that abstracts from this mechanically induced discontinuity.³² Because CEX income questions refer to the previous 12 months, we convert all income measures to quarterly amounts by dividing by four. We define private income as the sum of private income components and transfer income as the sum of transfer components. Table 5 lists the CEX variables used to construct private income and transfer income for three different periods, reflecting changes in CEX variable names over time. We keep only households for which this bottom-up sum matches the headline total income measure. The CEX records insurance reimbursements as negative expenditures and business losses as negative income. Because we work with logs, we drop households with non-positive private income, non-positive consumption, or outliers with quarterly consumption that exceeds 5 times of reported annual post-tax income.

Figure 9 presents the evolution of cross-quantile moments. The cross-quantile standard deviations of private income $\sigma_t^{(m)}$ are higher than those of post-tax income ($\sigma_t^{(d)}$) and consumption ($\sigma_t^{(c)}$). Moreover, $\sigma_t^{(d)}$ is larger than $\sigma_t^{(c)}$ in most periods. Standard deviation $\sigma_t^{(m)}$ exhibits substantial variation over time, increasing during the 2008 financial crisis and the COVID-19 period. In contrast, $\sigma_t^{(d)}$ and $\sigma_t^{(c)}$ are lower and vary less over time. Figure 10 shows the 90-10 percentile differences for all three variables. The difference is largest and most volatile for private income, smaller for post-tax income, and smallest for consumption.

³¹See [Improving Data Quality in the Consumer Expenditure Survey with TAXSIM](#), which documents the methodological change and its implications for measured tax liabilities and post-tax income.

³²[Heathcote et al. \(2010\)](#) also follow a similar splicing procedure when studying the time-series of hours worked across definitional changes to the data.

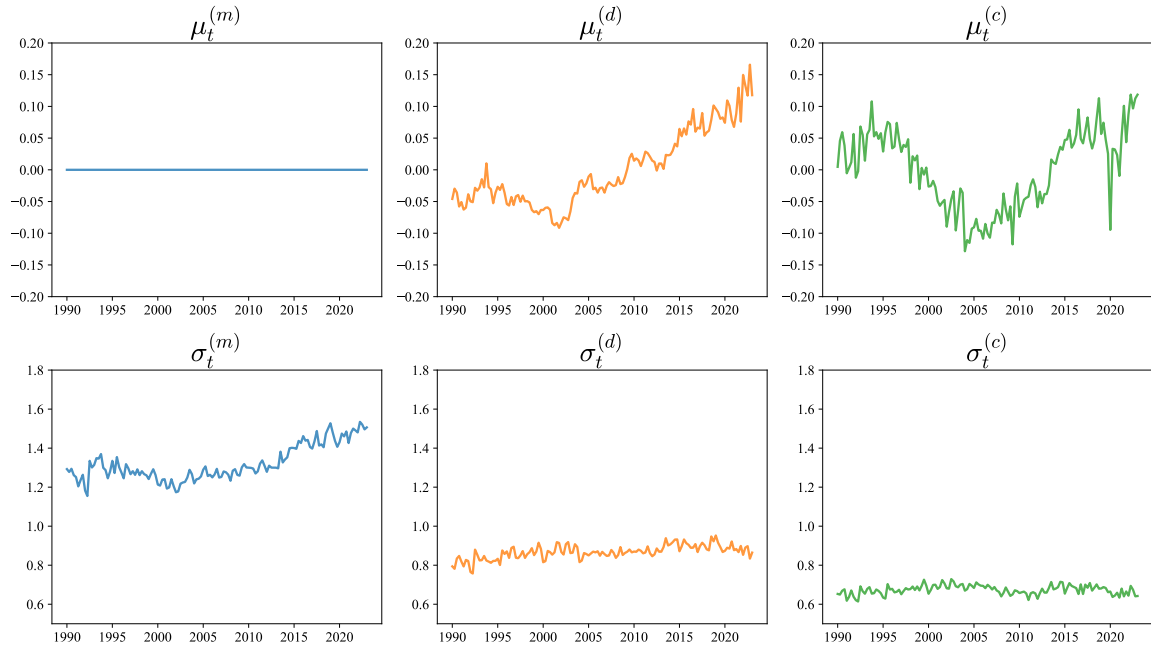


Figure 9: Cross-quantile moments

The cross-quantile standard deviation of private income has increased since 2008, but a similar trend is not apparent for post-tax income or consumption.

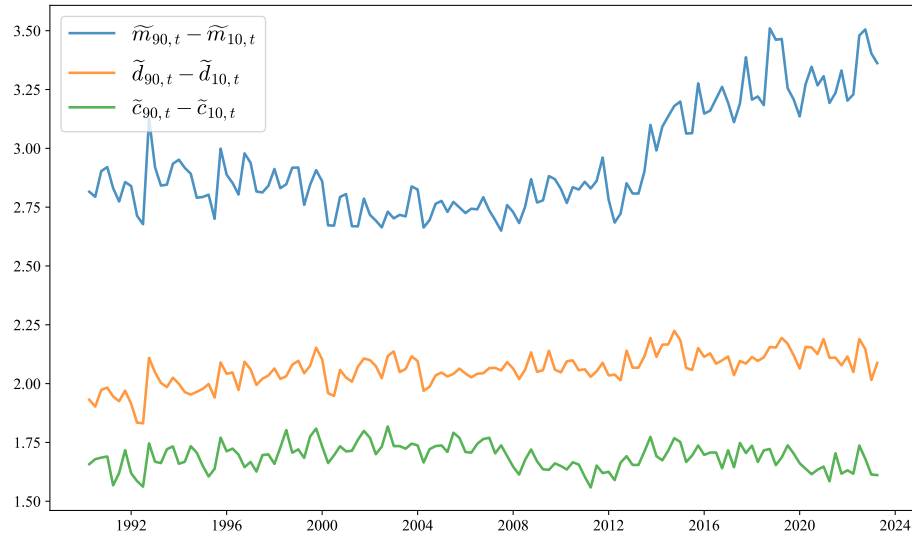


Figure 10: 90–10 percentile gap

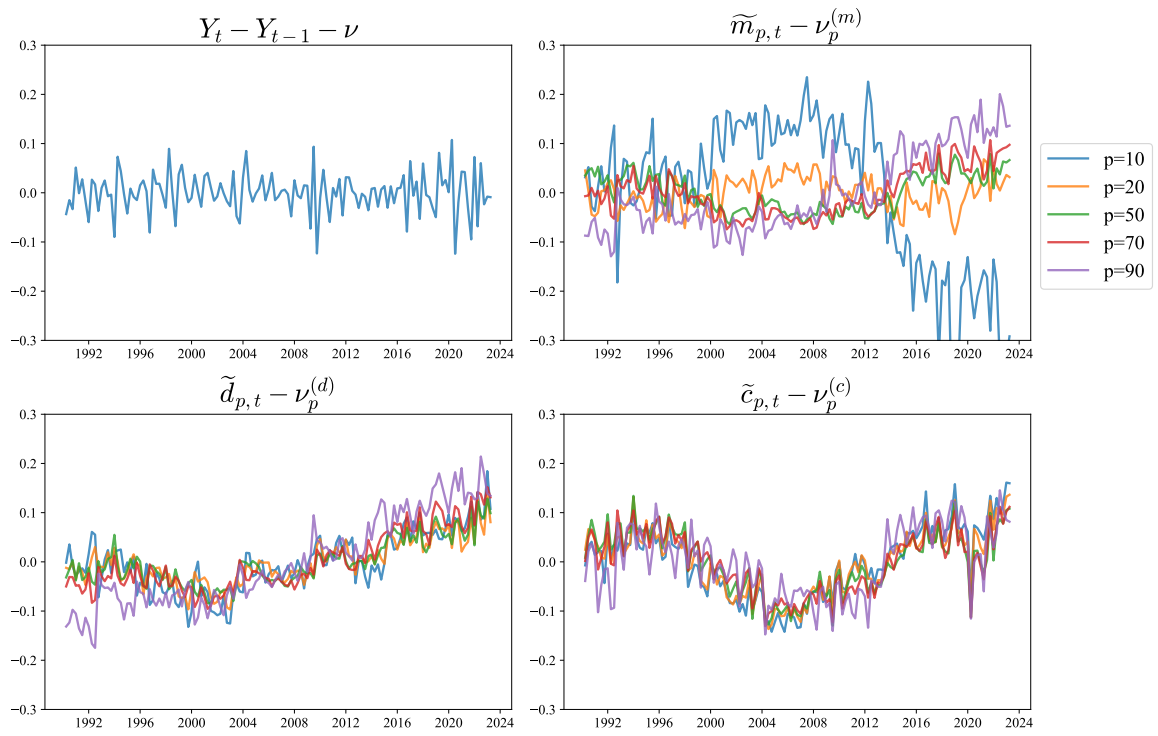


Figure 11: Demeaned quantiles, $\tilde{m}_{p,t}$, $\tilde{d}_{p,t}$, $\tilde{c}_{p,t}$

	Code Mnemonic		
	1990-2004	2004-2013	2013-2023
Private income			
Income from salary or wages	FSALARYX	FSALARYM	FSALARYM
Income from non-farm business	FNONFRMX	FNONFRMM	FSMPFRXM
Income from own farm	FFRMINCX	FFRMINCM	
Income from interest on savings accounts or bonds	INTEARNX	INTEARNM	INTRDVXM
Regular income earned from dividends, royalties, estates	FININCX	FININCXM	ROYESTXM
Income from pensions or annuities	PENSIONX	PENSIONM	RETSURVM
Net income or loss received from roomers or boarders	INCLOSSA	INCLOSAM	
Net income or loss received other rental properties	INCLOSSB	INCLOSBM	NETRENTM
Income from care of foster children, cash scholarships	OTHRINCX	OTHRINCM	OTHRINCM
Transfer income			
Income from Social Security benefits and Railroad Benefit checks	FRRETIRX	FRRETIRM	FRRETIRM
Supplemental Security Income from all sources	FSSIX	FSSIXM	FSSIXM
Income from unemployment compensation	UNEMPLX	UNEMPLXM	Combined in OTHREGXM
Income from workmen's compensation and veteran's payments	COMPENSX	COMPENSM	Combined in OTHREGXM
Income from public assistance including job training	WELFAREX	WELFAREM	WELFAREM
Income from other child support	CHDOTHX	CHDOTHXM	Combined in OTHREGXM
Income from regular contributions from alimony and other	ALIOTHX	ALIOTHXM	Combined in OTHREGXM
Food stamps	JFDSTMPA		
Food stamps and electronic benefits	FOODSMPX	FOODSMPM	JFSAMTM

Table 5: Categorizing CEX income into private and transfers

B Proofs

B.1 Proof of Proposition 1

Proof of Proposition 1. Let $\mathcal{F}_t := \sigma(\mathbf{y}_0, \mathbf{a}_1, \dots, \mathbf{a}_t)$. Under the maintained Gaussian VAR assumption, $\mathbb{E}[\mathbf{a}_{t+1} \mid \mathcal{F}_t] = \mathbf{0}$. Define $Z_t := Y_t + \mathbf{g} \mathbf{y}_t$. Using (6) and (7),

$$Z_{t+1} - Z_t - \nu = (Y_{t+1} - Y_t - \nu) + \mathbf{g}(\mathbf{y}_{t+1} - \mathbf{y}_t) = [e_1 \mathbf{B} + \mathbf{g}(\mathbf{B} - \mathbf{I})] \mathbf{y}_t + (e_1 + \mathbf{g}) \mathbf{a}_{t+1}.$$

Choose $\mathbf{g}$ so that $e_1 \mathbf{B} + \mathbf{g}(\mathbf{B} - \mathbf{I}) = \mathbf{0}$, equivalently $\mathbf{g}(\mathbf{I} - \mathbf{B}) = e_1 \mathbf{B}$, which yields $\mathbf{g} = e_1 \mathbf{B}(\mathbf{I} - \mathbf{B})^{-1}$ as in (10). With this choice, we have

$$Z_{t+1} - Z_t - \nu = \mathbf{H} \mathbf{a}_{t+1}, \quad \mathbf{H} := e_1 + \mathbf{g}.$$

Substituting the expression for $\mathbf{g}$ from (10) gives $\mathbf{H} = e_1 + e_1 \mathbf{B}(\mathbf{I} - \mathbf{B})^{-1}$ as in (9), and equivalently, $\mathbf{H} = e_1(\mathbf{I} - \mathbf{B})^{-1}$. Since $Z_{t+1} = Z_t + \nu + \mathbf{H} \mathbf{a}_{t+1}$, taking conditional expectations yields $\mathbb{E}[Z_{t+1} \mid \mathcal{F}_t] = Z_t + \nu$ and hence $\{Z_t - t\nu\}$ is a martingale with differences $\mathbf{H} \mathbf{a}_{t+1}$. Iterating gives

$$Z_t = Z_0 + t\nu + \sum_{j=1}^t \mathbf{H} \mathbf{a}_j.$$

Iterating (6) yields

$$\mathbf{y}_t = \mathbf{B}^t \mathbf{y}_0 + \sum_{j=0}^{t-1} \mathbf{B}^j \mathbf{a}_{t-j}.$$

Since $\rho(\mathbf{B}) < 1$, we have $\mathbf{B}^t \rightarrow \mathbf{0}$ and the infinite sum $\sum_{j=0}^{\infty} \mathbf{B}^j \mathbf{a}_{t-j}$ converges in L^2 , so $\{\mathbf{y}_t\}$ is an asymptotically stationary stochastic process and any fixed linear combination such as $\{\mathbf{g} \mathbf{y}_t\}$ is also asymptotically stationary. Using $Z_t = Y_t + \mathbf{g} \mathbf{y}_t$ and $Z_0 = Y_0 + \mathbf{g} \mathbf{y}_0$, rearranging gives

$$Y_t = t\nu + \sum_{j=1}^t \mathbf{H} \mathbf{a}_j - \mathbf{g} \mathbf{y}_t + (\mathbf{g} \mathbf{y}_0 + Y_0),$$

so $t\nu$ is a deterministic trend, $\sum_{j=1}^t \mathbf{H} \mathbf{a}_j$ is a martingale component, $-\mathbf{g} \mathbf{y}_t$ is an asymptotically stationary component, and $(\mathbf{g} \mathbf{y}_0 + Y_0)$ is a constant, which is (8). $\square$

B.2 Proof of Corollary 1

Proof of Corollary 1. We show this for the p th private-income quantile $m_{p,t}$. Define $e_p^{(m)} \in \mathbb{R}^{1 \times (3M+1)}$ to select the component of $\mathbf{y}_t$ corresponding to $\tilde{m}_{p,t} - \nu_p^{(m)}$. By definition, $\tilde{m}_{p,t} = m_{p,t} - Y_t$ and, from (5), $e_p^{(m)} \mathbf{y}_t = \tilde{m}_{p,t} - \nu_p^{(m)}$. Hence $m_{p,t} = Y_t + e_p^{(m)} \mathbf{y}_t + \nu_p^{(m)}$. Substituting (8) for Y_t yields

$$m_{p,t} = \underbrace{t\nu}_{\text{Trend}} + \underbrace{\sum_{j=1}^t \mathbf{H} \mathbf{a}_j}_{\text{Martingale}} - \underbrace{(\mathbf{g} - e_p^{(m)}) \mathbf{y}_t}_{\text{Stationary}} + \underbrace{(\mathbf{g} \mathbf{y}_0 + Y_0 + \nu_p^{(m)})}_{\text{Constant}}.$$

This has the form (11) with $u = m$, $e_p^{(u)} = e_p^{(m)}$, and $\nu_p^{(u)} = \nu_p^{(m)}$. Define analogously $e_p^{(d)}$ and $e_p^{(c)}$ that select components $\tilde{d}_{p,t} - \nu_p^{(d)}$ and $\tilde{c}_{p,t} - \nu_p^{(c)}$ from $\mathbf{y}_t$. The same argument yields (11) for $u = d$ and $u = c$. $\square$

B.3 Proof of Proposition 2

Proof of Proposition 2. Define $\bar{v}_p(\zeta, c_p) := v_p(\zeta, c_p) - c_p$. The recursion implies

$$\begin{aligned} \bar{v}_p(\zeta, c_p) &= \beta \mathbb{E} [\bar{v}_p(\zeta', c'_p) + (c'_p - c_p)] \\ &= \beta \mathbb{E} \bar{v}_p(\zeta', c'_p) + \beta \mathbf{D}_p^{(c)} \zeta, \end{aligned}$$

where the second equality uses (28). Conjecture $\bar{v}_p(\zeta, c_p) = \boldsymbol{\lambda}_p^{(c)\top} \zeta$ for some row vector $\boldsymbol{\lambda}_p^{(c)\top}$. Using $\mathbb{E} \zeta' = \mathbf{A} \zeta$ from (27), we obtain

$$\boldsymbol{\lambda}_p^{(c)\top} \zeta = \beta \boldsymbol{\lambda}_p^{(c)\top} \mathbf{A} \zeta + \beta \mathbf{D}_p^{(c)} \zeta.$$

Since this must hold for all ζ , we have $\boldsymbol{\lambda}_p^{(c)\top} (\mathbf{I} - \beta \mathbf{A}) = \beta \mathbf{D}_p^{(c)}$. Because $\mathbf{A} = \text{diag}(1, \mathbf{B})$ with $\rho(\mathbf{B}) < 1$ and $\beta \in (0, 1)$, we have $\rho(\beta \mathbf{A}) = \beta < 1$, so $(\mathbf{I} - \beta \mathbf{A})$ is invertible, yielding (30). Substituting back into $v_p(\zeta, c_p) = c_p + \bar{v}_p(\zeta, c_p)$ gives (29). $\square$

C DMD Algorithm

We use the following (DMD) algorithm to estimate $\hat{\mathbf{B}}$:

1. Construct data matrices

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \cdots & \mathbf{y}_{T-1} \end{bmatrix}, \quad \mathbf{Y}' = \begin{bmatrix} \mathbf{y}_2 & \mathbf{y}_3 & \cdots & \mathbf{y}_T \end{bmatrix}$$

where $\mathbf{Y}, \mathbf{Y}' \in \mathbb{R}^{(3M+1) \times (T-1)}$.

2. Compute the singular value decomposition (SVD)

$$\mathbf{Y} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top.$$

3. Retain the N leading singular values and vectors to form $\mathbf{U}_N \mathbf{\Sigma}_N \mathbf{V}_N^\top$, so that

$$\mathbf{Y} \approx \mathbf{U}_N \mathbf{\Sigma}_N \mathbf{V}_N^\top.$$

4. Compute a $(3M + 1) \times N$ matrix $\mathbf{\Phi}$ and an $N \times N$ diagonal matrix $\mathbf{\Lambda}$ with the following formulas:

$$\tilde{\mathbf{B}} = \mathbf{U}_N^\top \mathbf{Y}' \mathbf{V}_N \mathbf{\Sigma}_N^{-1}, \quad \tilde{\mathbf{B}} \mathbf{W} = \mathbf{W} \mathbf{\Lambda}, \quad \mathbf{\Phi} = \mathbf{Y}' \mathbf{V}_N \mathbf{\Sigma}_N^{-1} \mathbf{W},$$

with $\tilde{\mathbf{x}}_t = \mathbf{\Phi}^+ \mathbf{y}_t$ where $\mathbf{\Phi}^+$ is the Moore-Penrose pseudoinverse of $\mathbf{\Phi}$.

The rank N least squares estimator and its modal identities are

$$\hat{\mathbf{B}} = \mathbf{Y}' \mathbf{V}_N \mathbf{\Sigma}_N^{-1} \mathbf{U}_N^\top, \quad \hat{\mathbf{B}} \mathbf{\Phi} = \mathbf{\Phi} \mathbf{\Lambda}, \quad \hat{\mathbf{B}} = \mathbf{\Phi} \mathbf{\Lambda} \mathbf{\Phi}^+. \quad (54)$$

Here $\mathbf{\Lambda}$ contains the DMD eigenvalues and $\mathbf{\Phi}$ the associated exact DMD modes for the rank N estimator $\hat{\mathbf{B}}$.

D Singular Values and DMD Modes

In this Appendix we define DMD “modes”, then present visualizations of our estimates of them. We begin by noting that [Sargent et al. \(2025\)](#) show that affiliated with VAR representation (12) is another representation that is cast in terms of the eigenvalues $\mathbf{\Lambda}$ and the DMD

modes Φ :

$$\hat{\mathbf{x}}_{t+1} = \Lambda \hat{\mathbf{x}}_t + \Lambda \Phi^+ \hat{\mathbf{a}}_t, \quad (55)$$

$$\mathbf{y}_t = \Phi \hat{\mathbf{x}}_t + \hat{\mathbf{a}}_t, \quad \hat{\mathbf{a}}_t \perp \mathbf{y}_{t-1}. \quad (56)$$

where

$$\hat{\mathbf{x}}_t = \mathbb{E}[\mathbf{x}_t \mid \mathbf{y}_{t-1}],$$

and $\mathbf{x}_t$ is the state vector in the following no-observation noise time-invariant LQG state-space representation

$$\begin{aligned} \mathbf{x}_{t+1} &= \underset{N \times N}{\mathbf{A}} \mathbf{x}_t + \underset{N \times N}{\mathbf{C}} \mathbf{w}_{t+1} \\ \mathbf{y}_t &= \underset{(3M+1) \times N}{\mathbf{G}} \mathbf{x}_t, \end{aligned}$$

where random shocks $\mathbf{w}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$ and

$$\mathbf{A} = \Lambda, \quad \mathbf{G} = \Phi. \quad (57)$$

In Figure 5, we plot singular values of the $\mathbf{Y}$ matrix. We decided to retain only the three largest singular values when we estimated $\hat{\mathbf{B}}$ via DMD. This choice was guided by a standard “elbow rule”: the singular values decline rapidly for the first few components and then flatten out, with a visible kink at the third singular value. We estimated:

$$\Lambda = \begin{bmatrix} 0.9629 & 0 & 0 \\ 0 & 0.8649 & 0 \\ 0 & 0 & 0.5472 \end{bmatrix} \quad (58)$$

Figure 12 plots time series of the DMD modes that we recovered using this estimate of Λ and the affiliated Φ . We associate each mode $\hat{x}_{n,t}$, $n \in \{1, 2, 3\}$, with the univariate time series of cross-sectional moments (means μ_t or standard deviations σ_t) with the highest

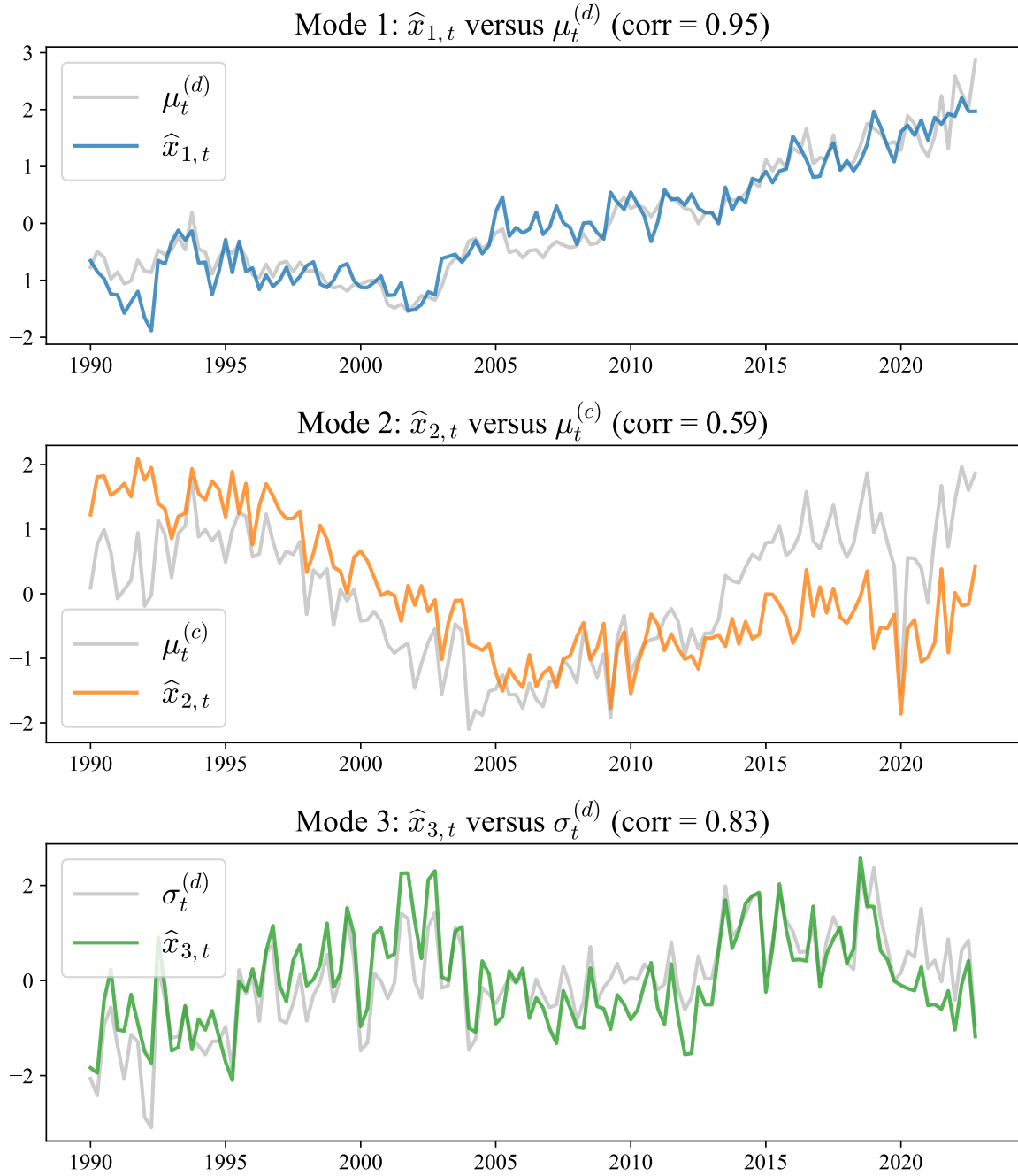


Figure 12: DMD modes and data moments

correlations.³³

Figure 12 associates the first three DMD factors with time variation in cross-quantile moments. After orienting each mode so that its realization is positively correlated with the

³³Cross-sectional moments are defined in Section 3.2.

associated moment, we find that $\hat{x}_{1,t}$ tracks the cross-quantile mean of post-tax income $\mu_t^{(d)}$ (corr. ≈ 0.95), $\hat{x}_{2,t}$ tracks the cross-quantile mean of consumption $\mu_t^{(c)}$ (corr. ≈ 0.59), and $\hat{x}_{3,t}$ tracks the cross-quantile standard deviation of post-tax income $\sigma_t^{(d)}$ (corr. ≈ 0.83).³⁴

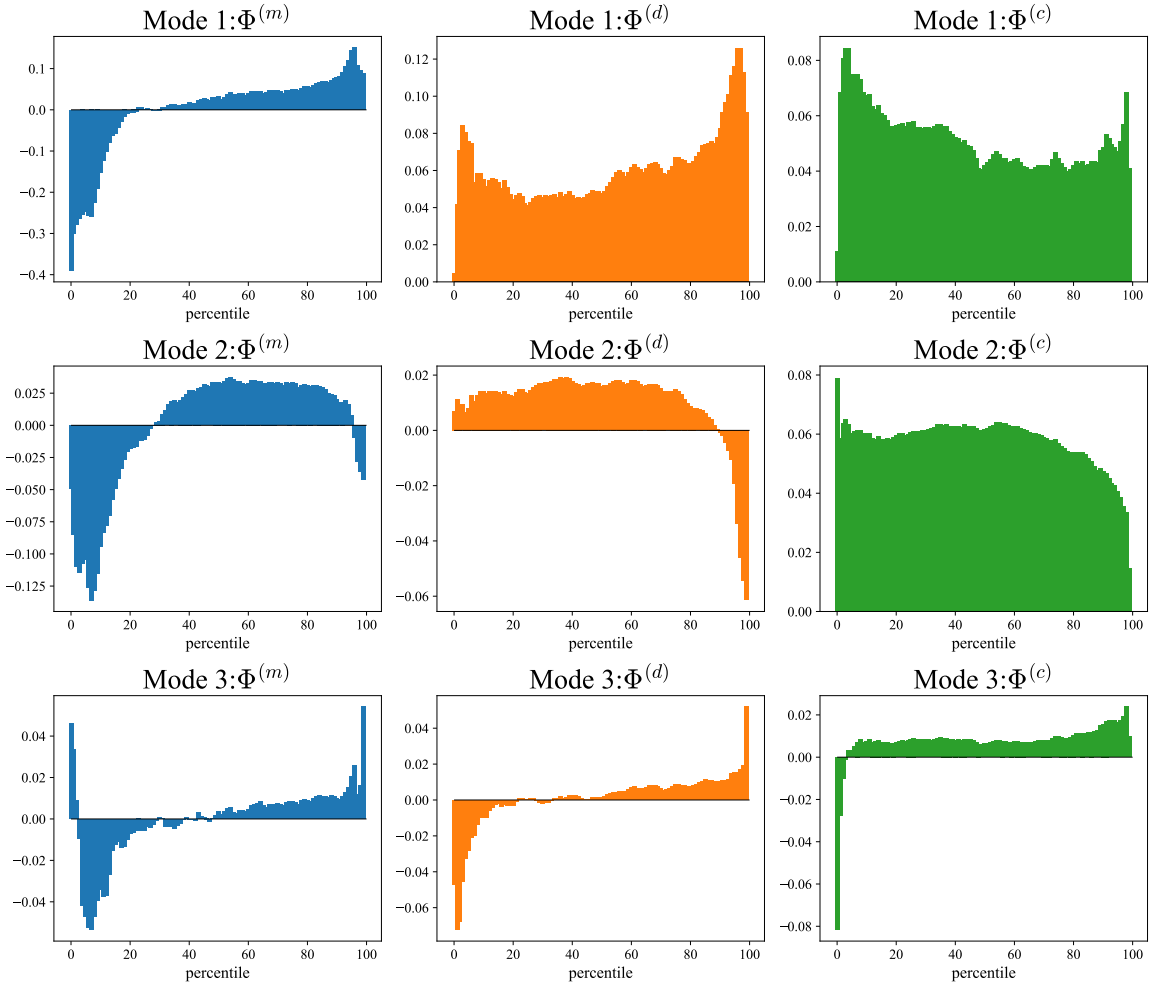


Figure 13: DMD loadings on modes 1–3

Mode 1 is strongly associated with the cross-quantile mean of post-tax income $\mu_t^{(d)}$. Since $\mu_t^{(m)} = 0$ for all t by construction, the loadings of private income quantiles must average to zero, thus shedding light on increases in income inequality associated with an increase in mode 1. The private-income loadings in Figure 13 exhibit a classic inequality pattern: low percentiles load negatively while high percentiles load positively, so increases in $\hat{x}_{1,t}$ widen

³⁴Modes are linked to the data moments by picking the time series of moments that have the highest absolute value of correlation with each mode. The absolute value is used because DMD modes and their amplitudes are only defined up to an arbitrary nonzero scaling.

the private-income distribution. In contrast, post-tax-income and consumption loadings on mode 1 are positive throughout the distribution, with relatively large responses in both tails. Thus, when mode 1 rises, the tax-and-transfer system partially offsets the adverse exposure of the lower tail of private income, and low-percentile consumption moves closely with low-percentile post-tax income, consistent with limited scope for consumption smoothing through private savings at the bottom.

Mode 2 is associated with the cross-quantile mean of consumption $\mu_t^{(c)}$. Its private-income loadings are negative for low percentiles and positive for middle percentiles. Post-tax-income loadings on mode 2 are small for most percentiles but turn negative in the upper tail. Consumption loadings on mode 2 are positive throughout the distribution, indicating that increases in $\hat{x}_{2,t}$ raise consumption broadly, with the richest households showing slightly smaller responses.

Mode 3 is associated with the cross-quantile standard deviation of post-tax income $\sigma_t^{(d)}$. Its loadings show notable variation at the tails of the distribution, with the extreme lower tail of consumption showing pronounced negative loadings. This mode captures time variation in post-tax income dispersion that is driven mainly by tail movements rather than broad-based changes across the entire distribution.

E VAR(1) Residual Diagnostics

Our VAR(1) specification (6) assumes that the innovation sequence $\{\mathbf{a}_t\}$ is serially uncorrelated. With $n = 301$ state variables and $T = 133$ observations, standard multivariate residual tests face severe rank deficiency. We address this using ridge regression, which regularizes the high-dimensional VAR coefficient matrix.

We fit ridge VAR(p) models on the residuals $\{\mathbf{a}_t\}$ and evaluate predictability using time-series cross-validation with $K = 5$ folds. For each lag order p , we select the ridge penalty parameter α via grid search over $\alpha \in \{10^{-4}, \dots, 10^4\}$, a set of 30 logarithmically-spaced values. The grid search chooses the α that maximizes the mean out-of-sample R^2 across folds. Within each training fold, lagged regressors are standardized to ensure scale-invariant penalty selection. If the VAR(1) innovation assumption holds, lagged residuals should have no predictive power for future residuals, yielding OOS, i.e., out-of-sample, $R^2 \approx 0$ or

negative.

Table 6 reports the results for lag orders $p = 1, \dots, 8$. The OOS R^2 values are close to zero and uniformly negative (ranging from about -0.11 to -0.05), confirming that lagged residuals have no meaningful predictive power, consistent with approximate serial uncorrelation. The highest OOS R^2 occurs at $p = 1$, and adding lags does not improve predictability.

Lag order p	OOS R^2	In-sample R^2	Best α
1	-0.0220	0.3446	0.7279
2	-0.0555	0.2818	4.8939
3	-0.0658	0.2722	9.2367
4	-0.0137	0.4279	4.8939
5	-0.0153	0.4721	4.8939
6	-0.0171	0.4320	9.2367
7	-0.0493	0.4508	9.2367
8	-0.0583	0.3757	17.4333

Table 6: Out-of-sample and in-sample R^2 for ridge VAR(p) on residuals.

One interesting observation is that the OOS R^2 values show a slight uptick at lag 4, suggesting quarterly seasonality in the residuals, which is later confirmed by the autocorrelation analysis. We relegate further discussion of this to the autocorrelation analysis below.

Figure 14 displays the coefficient distributions from ridge VAR(p) fits to the residuals $\{\mathbf{a}_t\}$ (left: $p = 1$; right: $p = 1, \dots, 8$). The coefficient distributions are concentrated around zero. For $p = 1$, the mean absolute coefficient is 0.0058, the standard deviation of coefficients is 0.0099, the maximum absolute coefficient is 0.2055, and 85.7% of coefficients have absolute value < 0.01 .

For the autocorrelation analysis, we compute the univariate ACF of each residual series and summarize them across the full set of percentiles. The ACF plot shows that mean and median autocorrelations are small and lie within the 95% confidence bands for white noise at short lags, with the largest deviation at lag 4. This pattern is consistent with the presence of quarterly seasonality that we deliberately did not remove because seasonal adjustment filters can distort cross-sectional dispersion and complicate the interpretation of innovations (see, e.g., Hansen and Sargent (1993) and Sims (1993)). Together, these results suggest that the VAR(1) innovation assumption is not seriously violated.

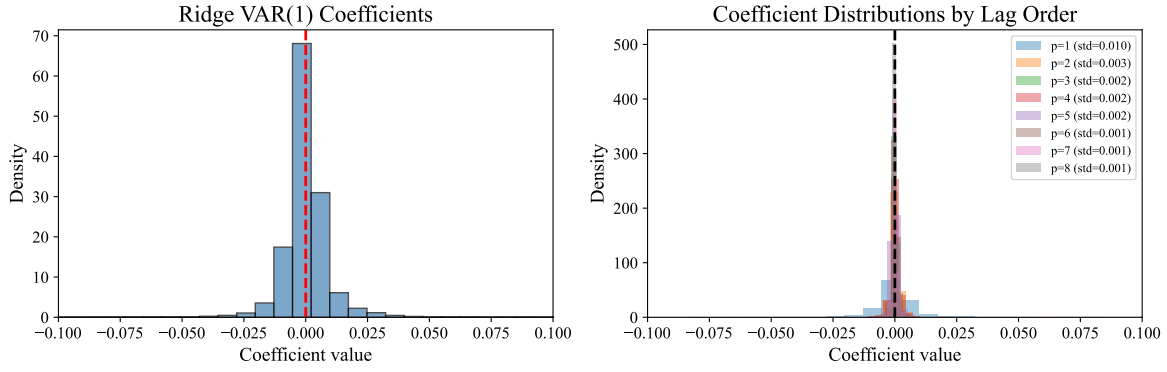


Figure 14: Coefficient distributions for ridge $\text{VAR}(p)$ fits to the residuals (left: $p = 1$; right: $p = 1, \dots, 8$).

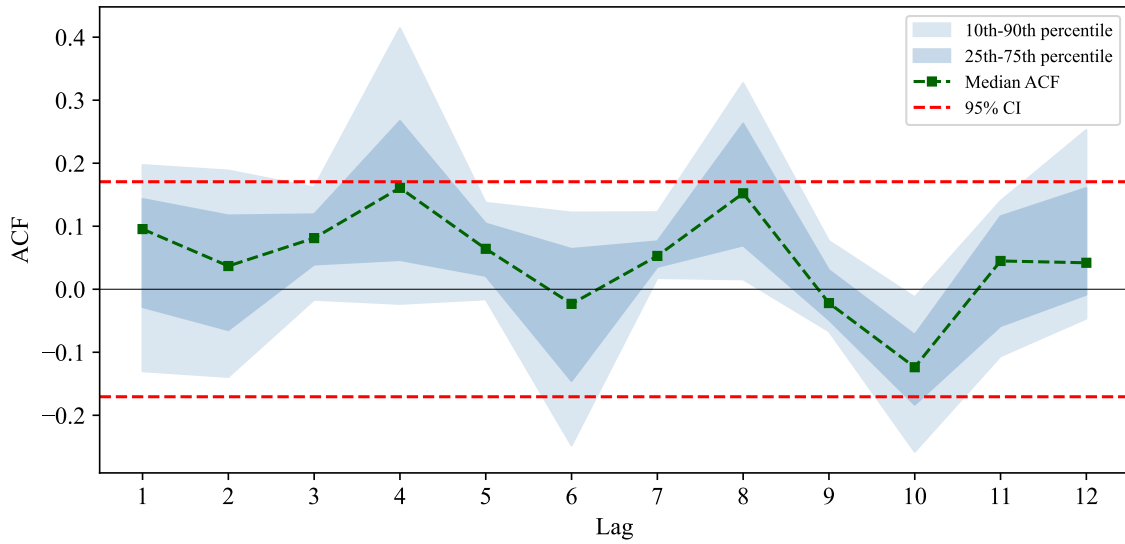


Figure 15: Autocorrelation functions of full residual series with 95% bands computed across components of $\mathbf{a}_t$.

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